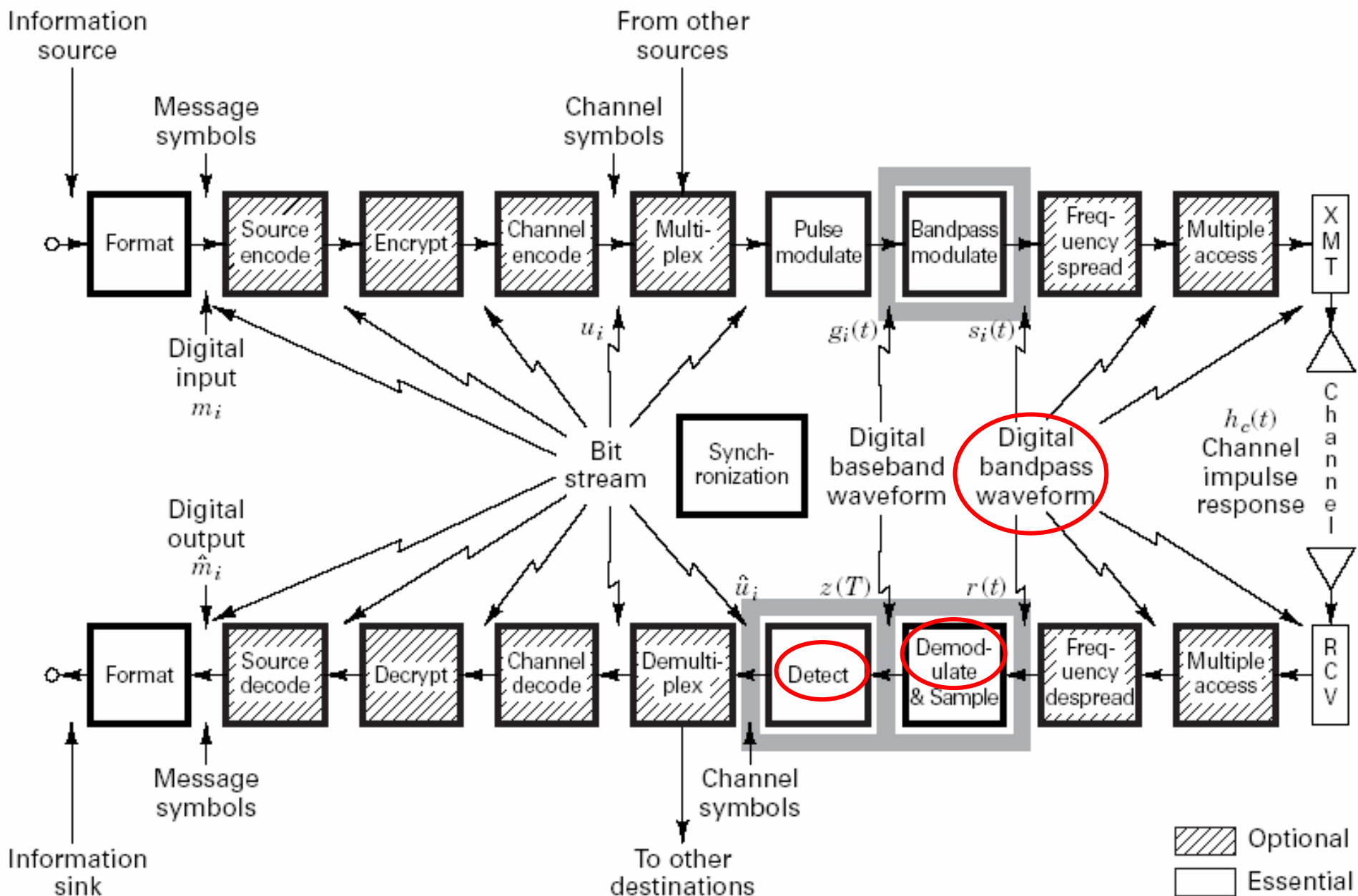


Passband Digital Transmission

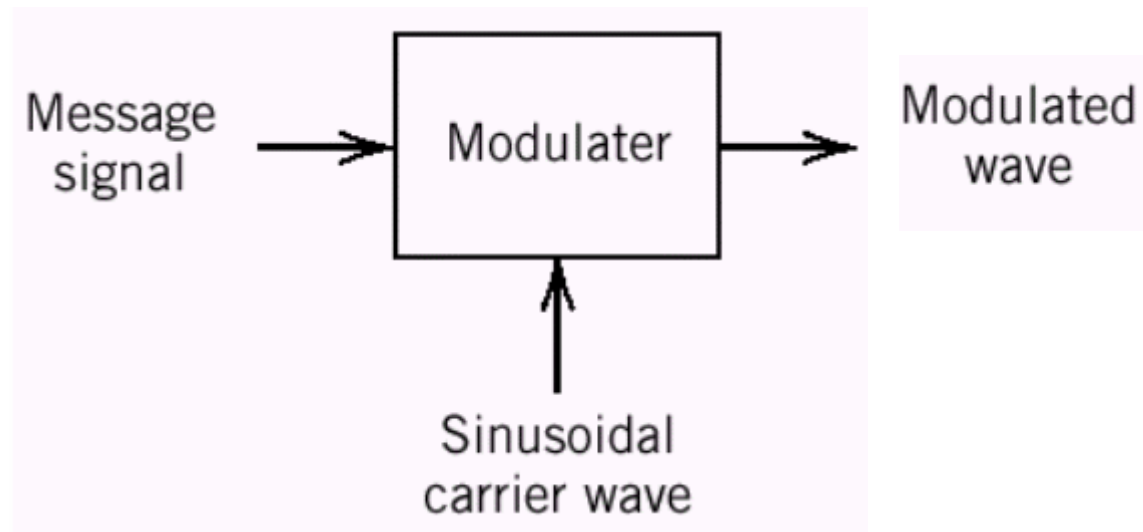


Contents

- **Digital Modulation**
 - *Binary* ASK, FSK, PSK, DPSK
 - *M-ary* ASK, FSK, PSK, DPSK
- **Sophisticated Digital Modulation**
CAP, DMT, MSK, GMSK...
- **Synchronization**

Digital Modulation

- Carrier • $\cos(2 \pi f_c t)$
- *Modulating Signal* • Baseband digital Signals 0,1...
- *Modulated Signal* • $s(t)$ → *Keying Signal*



Signal Space & Constellation

Signal Space

- Signal Vector: s_i ($i=1,2,\dots,M$)

$$s_i = \begin{bmatrix} s_{i1} \\ s_{i2} \\ \vdots \\ s_{iN} \end{bmatrix}$$

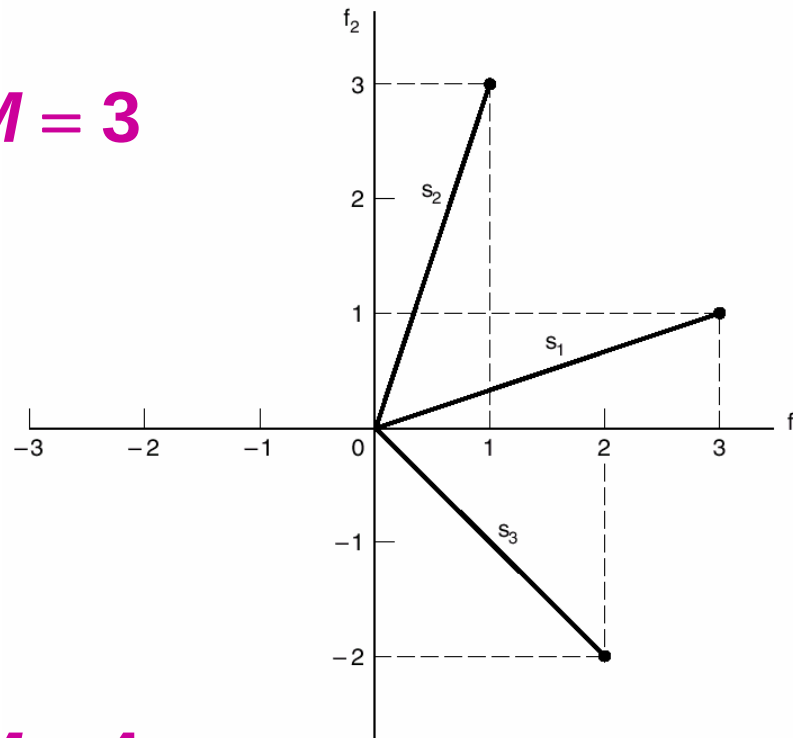
$$N \leq M$$

- **Signal Space:**
N-dimensional Euclidean Space:
N mutually perpendicular axes
labeled $\Phi_1, \Phi_2, \dots, \Phi_N$

Geometric Representation

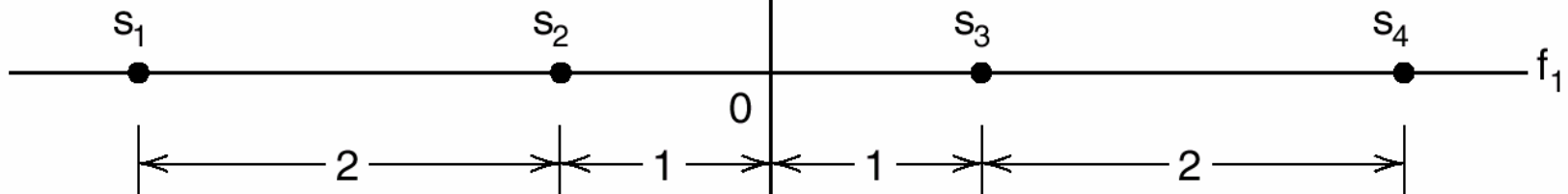
$N = 2$ and $M = 3$

$N \leq M$

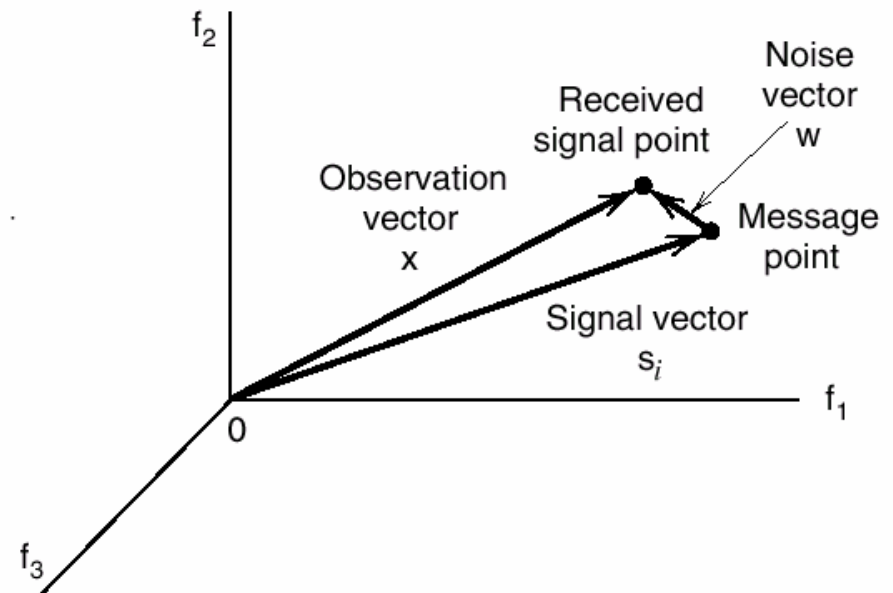
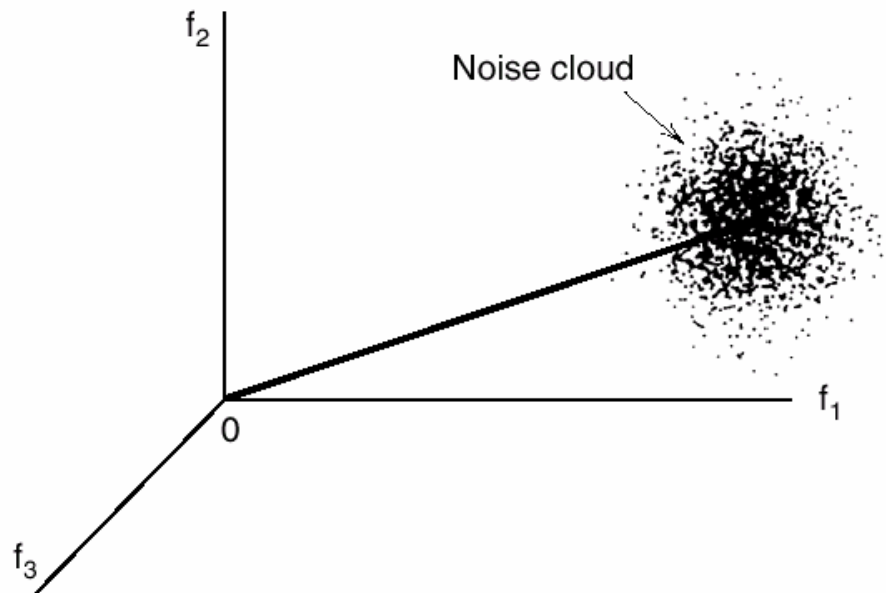


$N = 1$ and $M = 4$:

2B1Q Code



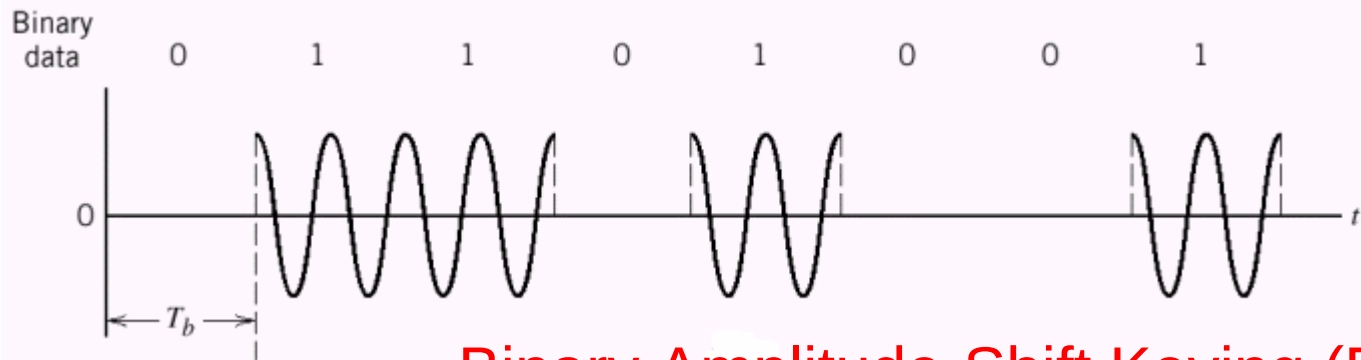
Constellation



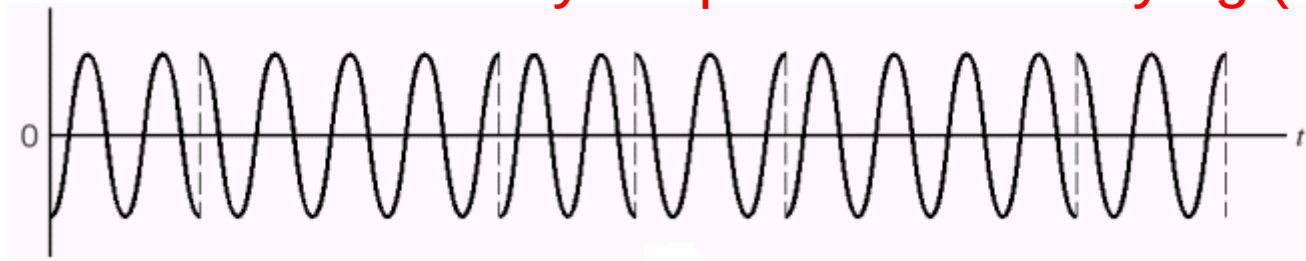
- Message Point = Transmitted Signal Point
- *Signal Vector s_i*
- Received Signal Point
- *Observation Vector x*

ASK, PSK(DPSK), FSK

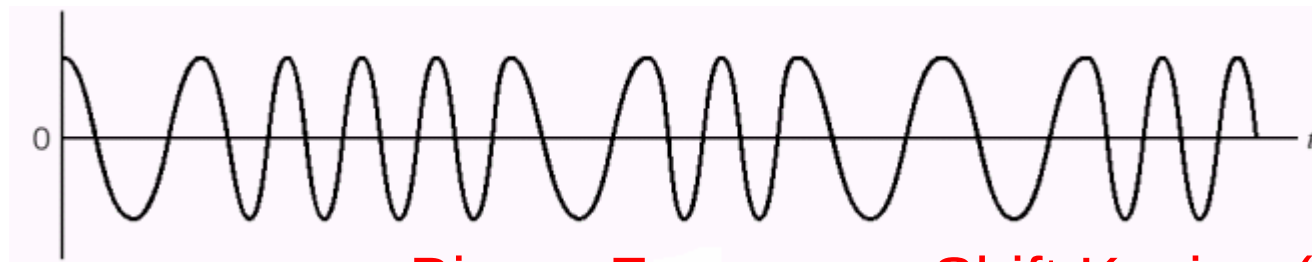
Introduction



- Binary Amplitude-Shift Keying (BASK)



- Binary Phase-Shift Keying (BPSK)



- Binary Frequency-Shift Keying (BFSK)

Binary Amplitude-Shift Keying

- BASK** Definition

$$\left\{ \begin{array}{l} s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) \\ s_2(t) = 0 \end{array} \right.$$

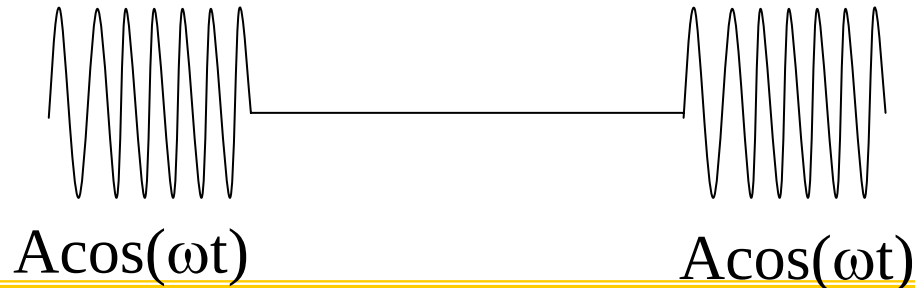
“1”

“0”

Baseband
Data



ASK
modulated
signal



Bandpass Modulation

Binary Phase-Shift Keying

- **BPSK** Definition

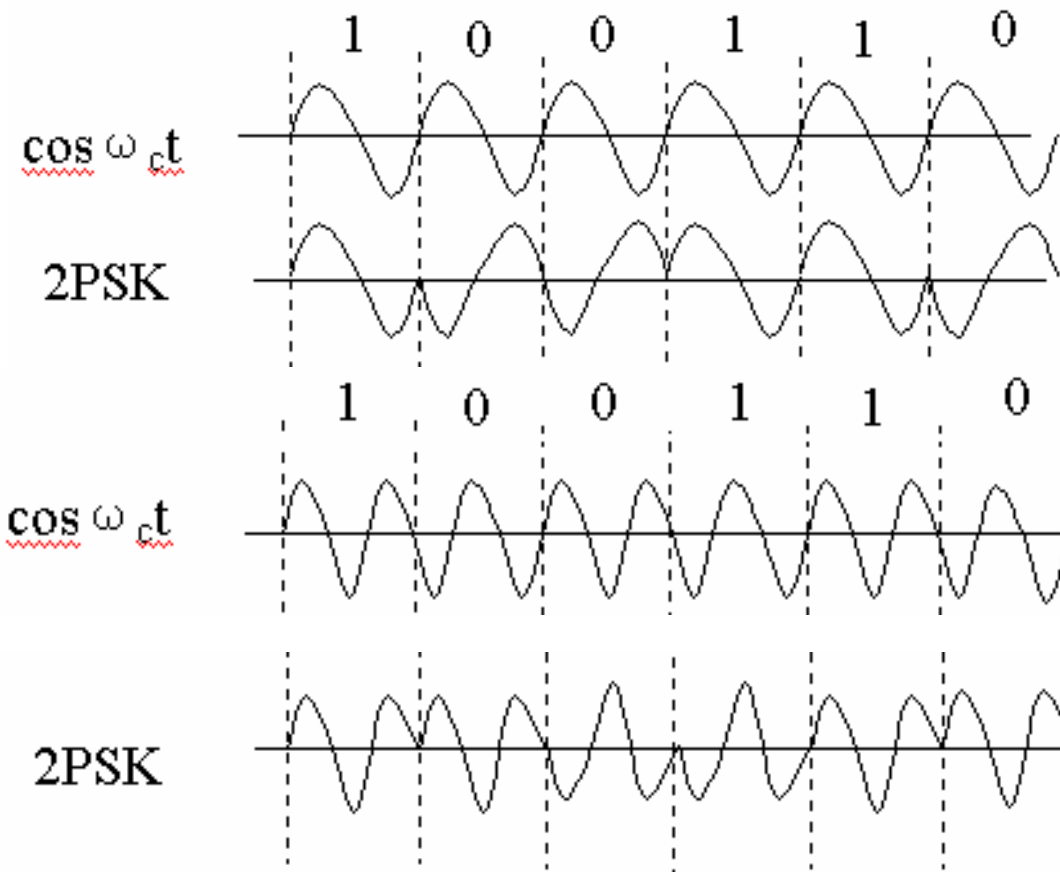
$$\left\{ \begin{array}{ll} s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) & \text{"1"} \\ s_2(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \pi) = -\sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) & \text{"0"} \end{array} \right.$$

- BPSK Waveform

compare to the phase of carrier

Binary Phase-Shift Keying(2)

- BPSK** Examples



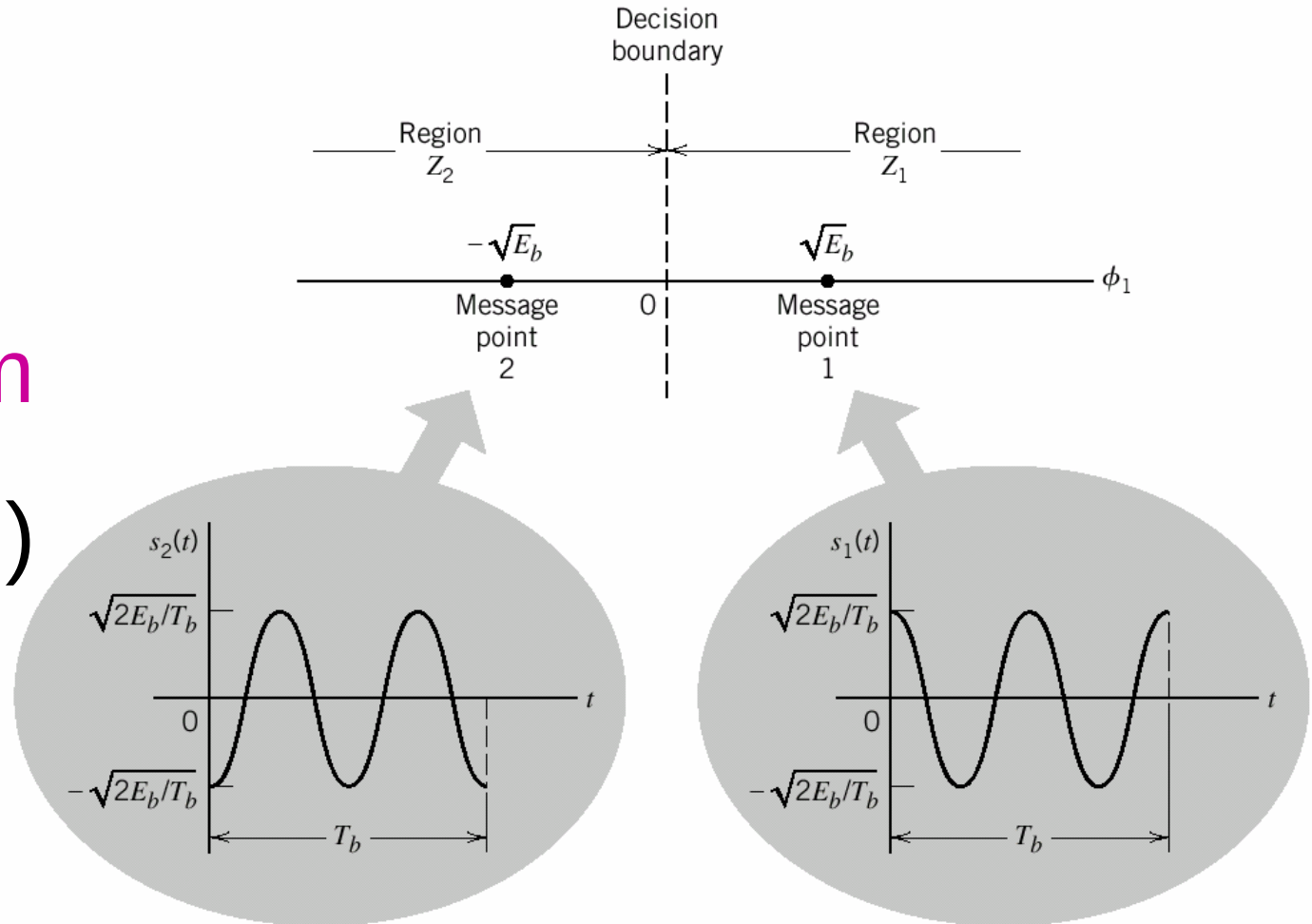
$$f_c = R_b$$

$$f_c = 1.5R_b$$

Binary Phase-Shift Keying(3)

BPSK Signal Space Diagram

$$(f_c = 2R_b)$$



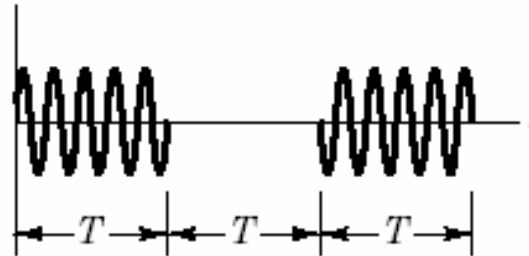
2ASK vs. 2PSK

ASK

$$s_i(t) = \sqrt{\frac{2E_i(t)}{T}} \cos(\omega_0 t + \phi)$$

$$i = 1, 2, \dots, M$$

$$0 \leq t \leq T$$

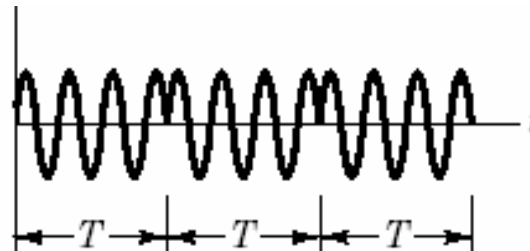


PSK

$$s_i(t) = \sqrt{\frac{2E}{T}} \cos(\omega_0 t + 2\pi i/M)$$

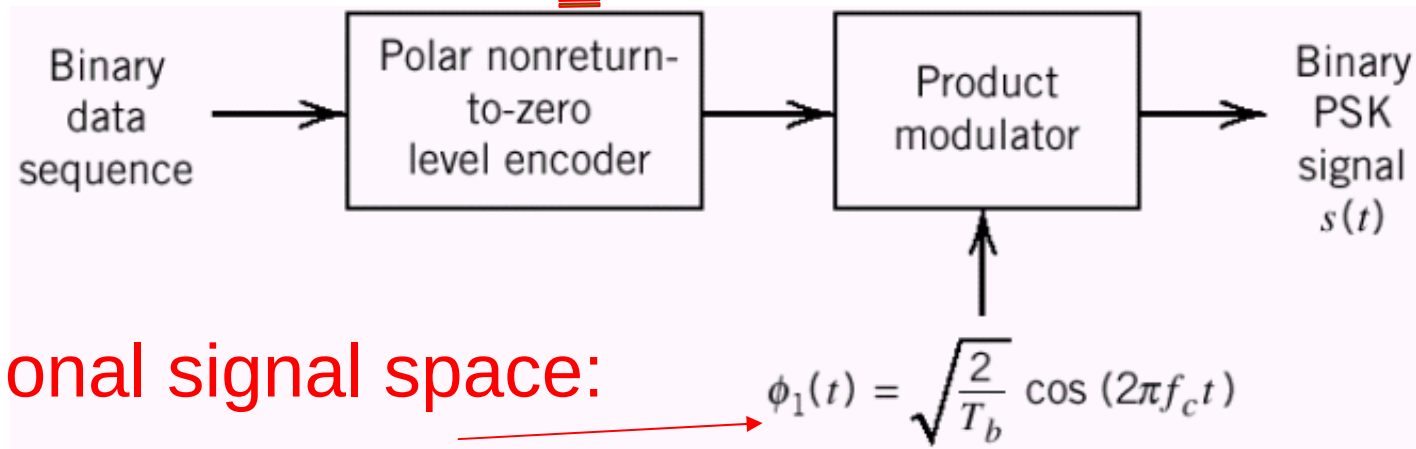
$$i = 1, 2, \dots, M$$

$$0 \leq t \leq T$$



Generation of BPSK

$$\left\{ \begin{array}{ll} s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) & \text{"0"} \rightarrow \sqrt{E_b} \\ s_2(t) = -\sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) & \text{"1"} \rightarrow -\sqrt{E_b} \end{array} \right.$$

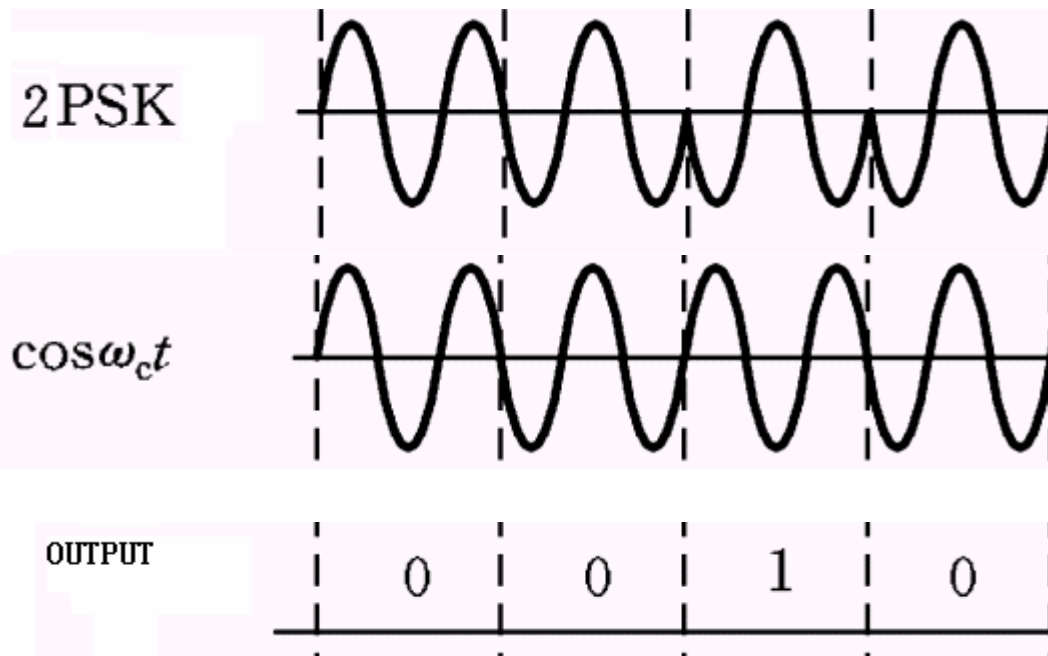


- One-dimensional signal space:

$$\phi_1(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_c t)$$

Detection of BPSK

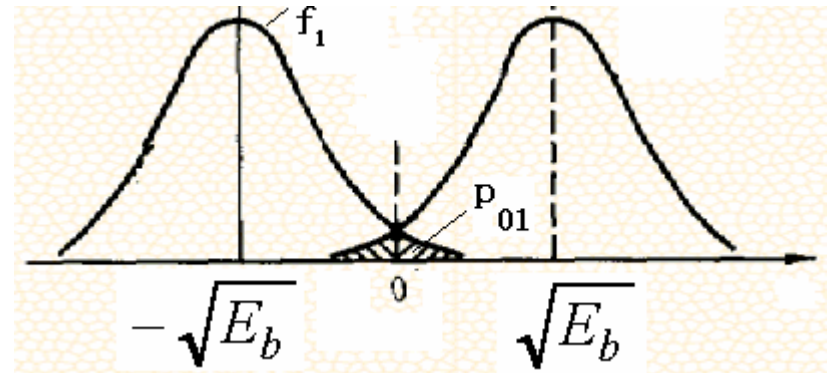
- BPSK: Compare to the phase of *carrier*



Assume: $\begin{cases} s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) & \text{"0"} \\ s_2(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \pi) & \text{"1"} \end{cases}$

P_e of BPSK

$$p_{01} = \int_0^{\infty} f_X(x|1)dx$$
$$= \frac{1}{\sqrt{\pi N_0}} \int_0^{\infty} e^{-\frac{1}{N_0}(x+\sqrt{E_b})^2} dx$$



$$= \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

erfc: Complementary Error Function

$$P_e = p_1 p_{01} + p_0 p_{10} = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

E_b : signal energy per bit

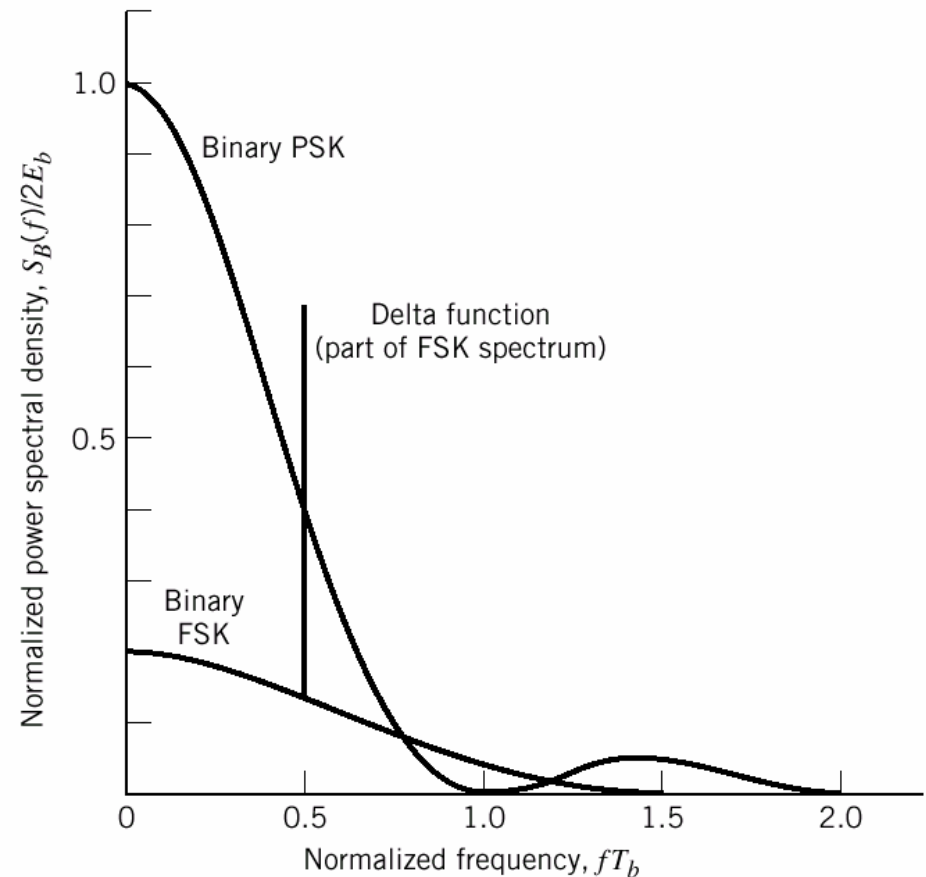
N_0 : noise spectral density

Power Spectra of BPSK

- The **baseband** **power spectral density** $S_B(f)$

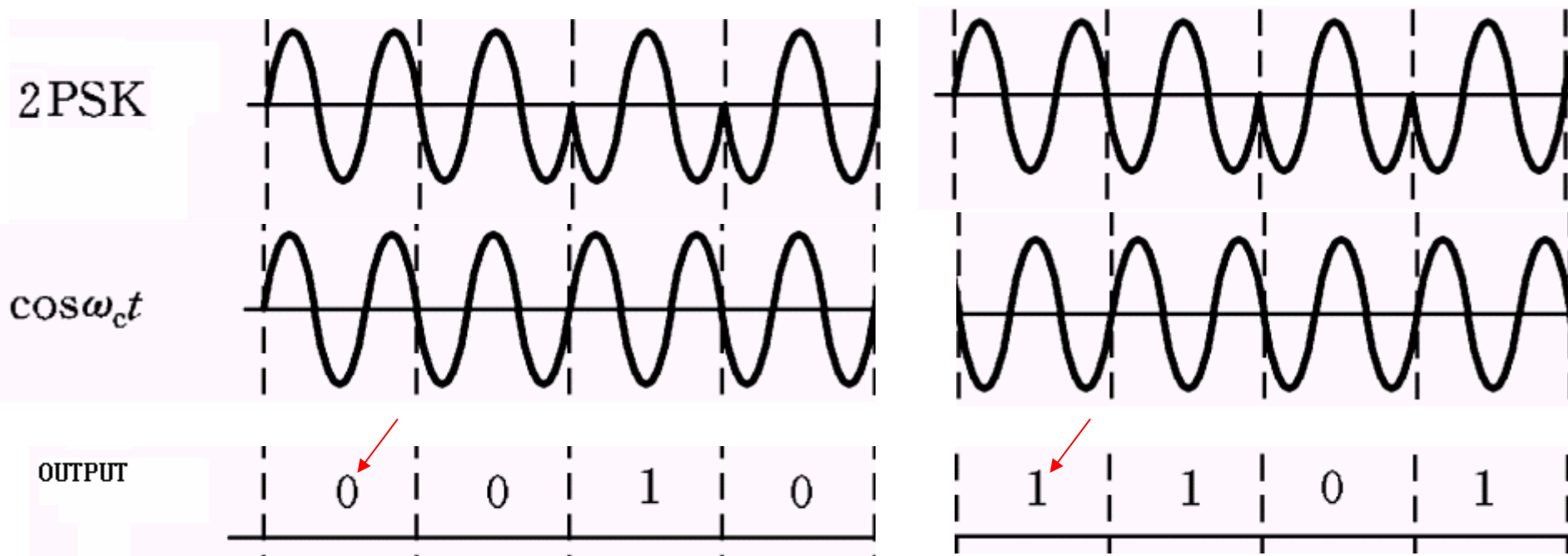
$$S_B(f) = \frac{2E_b \sin^2(\pi T_b f)}{(\pi T_b f)^2}$$

The power spectrum falls off as the inverse square of frequency



Drawback of PSK Detection

- BPSK: Compare to the phase of *carrier*

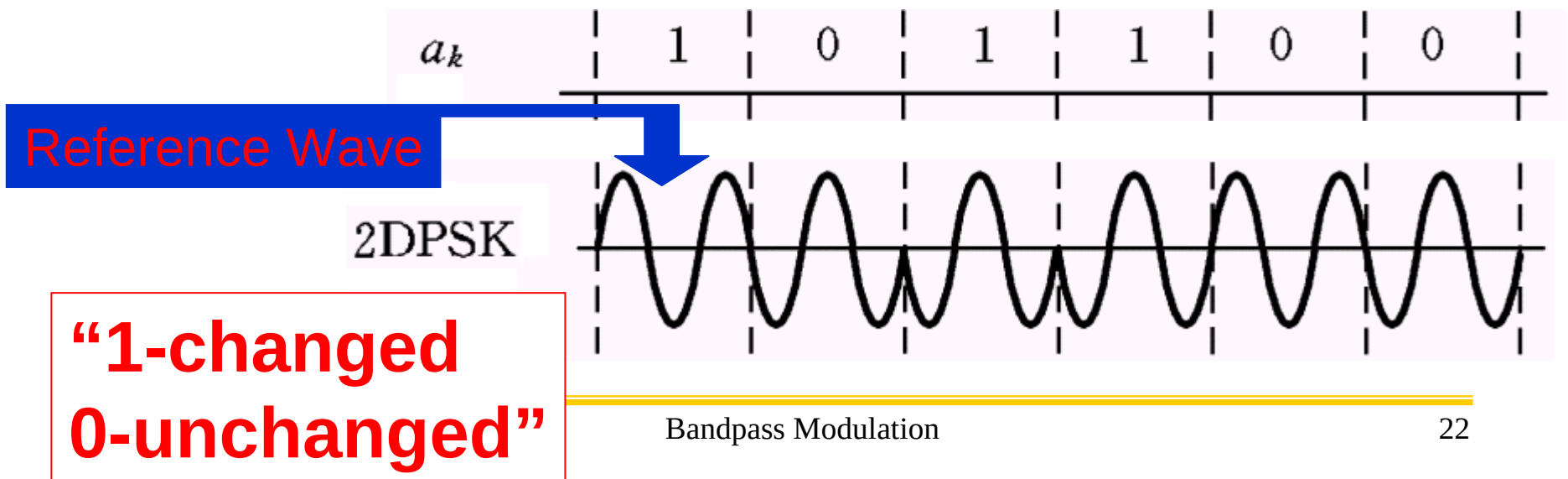


Assume:

$$\begin{cases} s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) & \text{"0"} \\ s_2(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \pi) & \text{"1"} \end{cases}$$

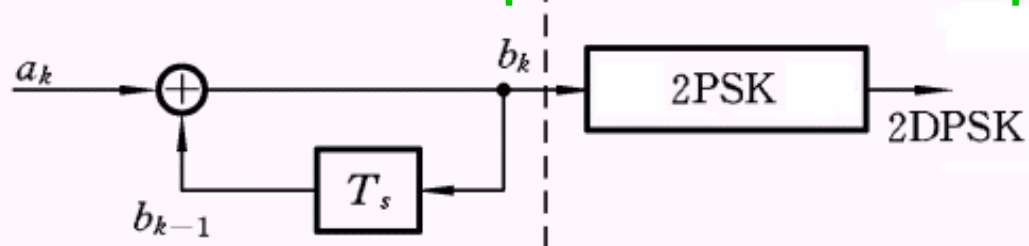
DPSK(Differential PSK)

- PSK: Compare to the phase of *carrier*
 - DPSK: Compare to its *own* previous phase
- { Adjacent phase changed π “1”
Adjacent phase unchanged “0”



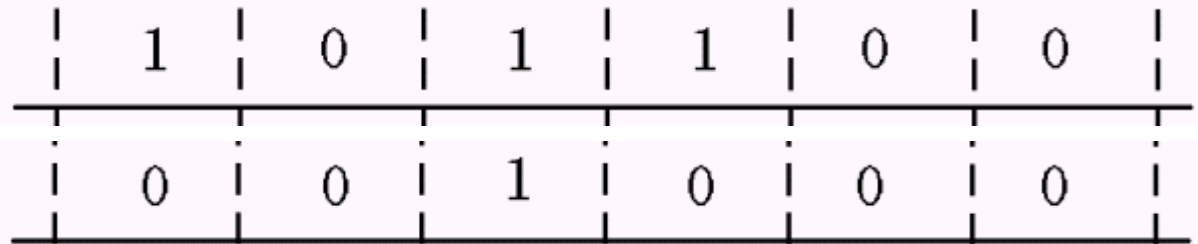
Generation of DPSK

- DPSK: Compare to its *own* previous phase



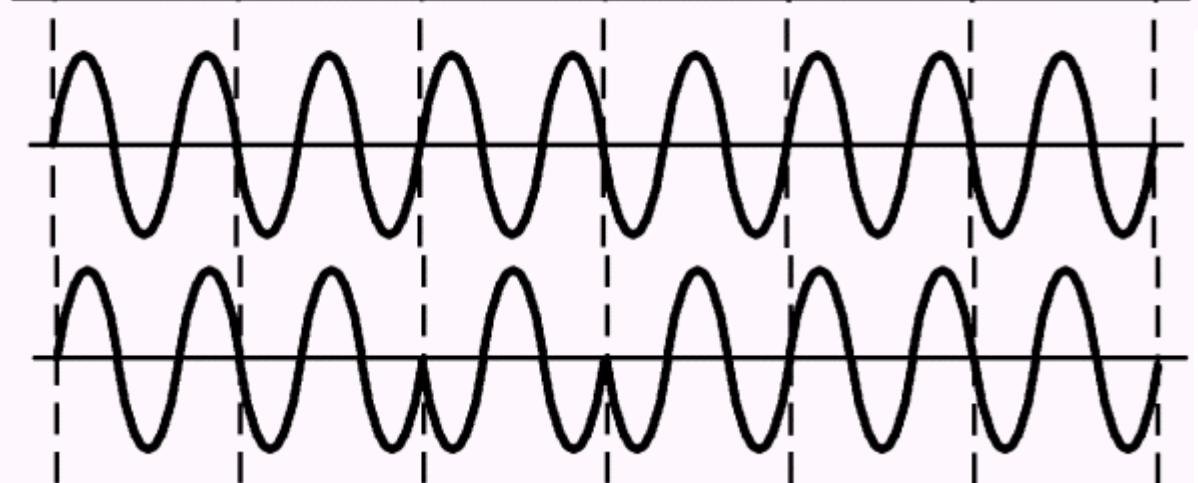
2DPSK
= Differential Code's 2PSK

a_k



Differential Code: b_k

$\cos \omega_c t$



2DPSK

QPSK

QPSK(Quadri-PSK)

- Purpose:

To increase the *bandwidth efficiency*

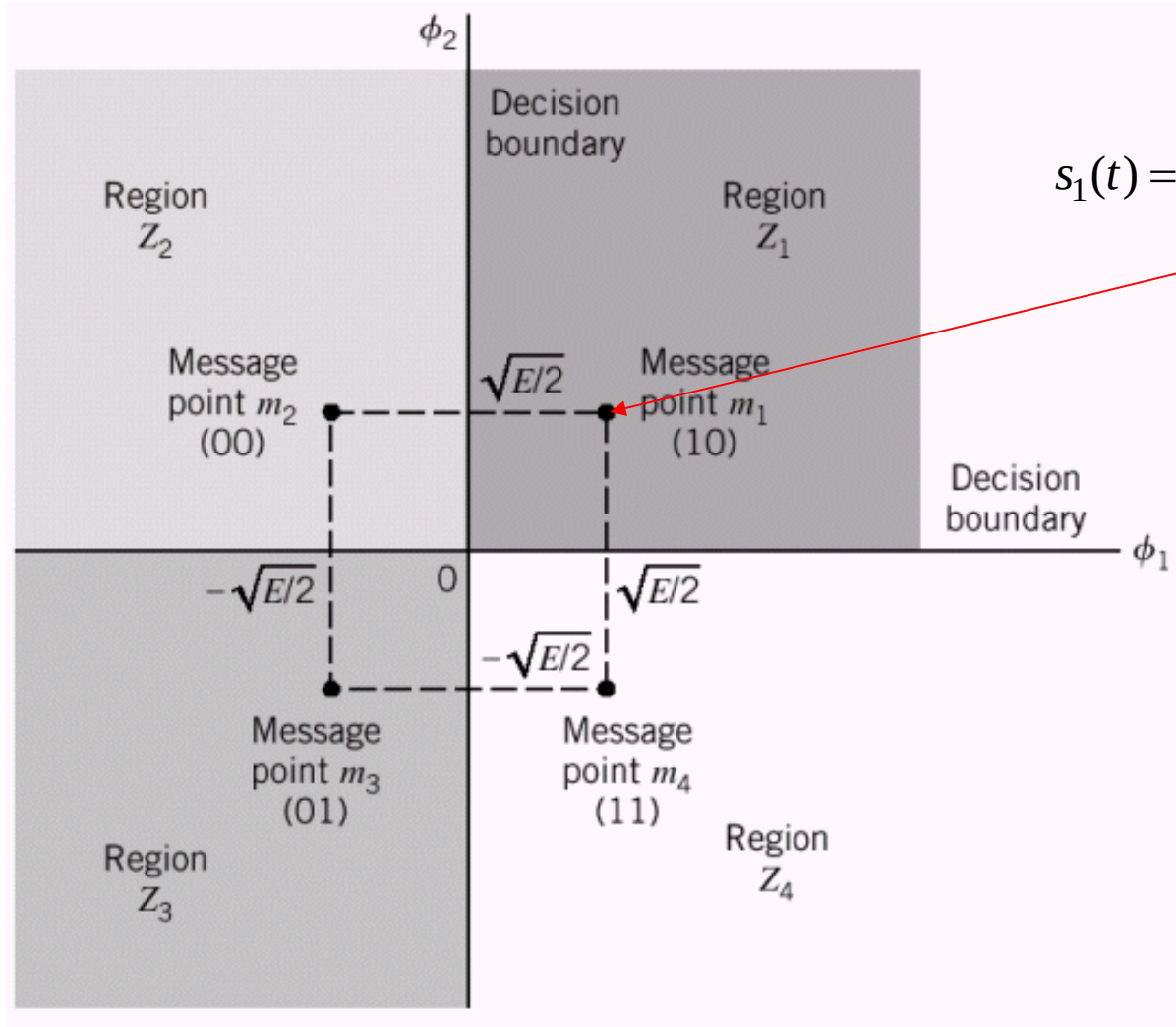
- The signal is $s_i(t)$:

$s_1(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \frac{\pi}{4})$	“10”
$s_2(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \frac{3\pi}{4})$	“00”
$s_3(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \frac{5\pi}{4})$	“01”
$s_4(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \frac{7\pi}{4})$	“11”

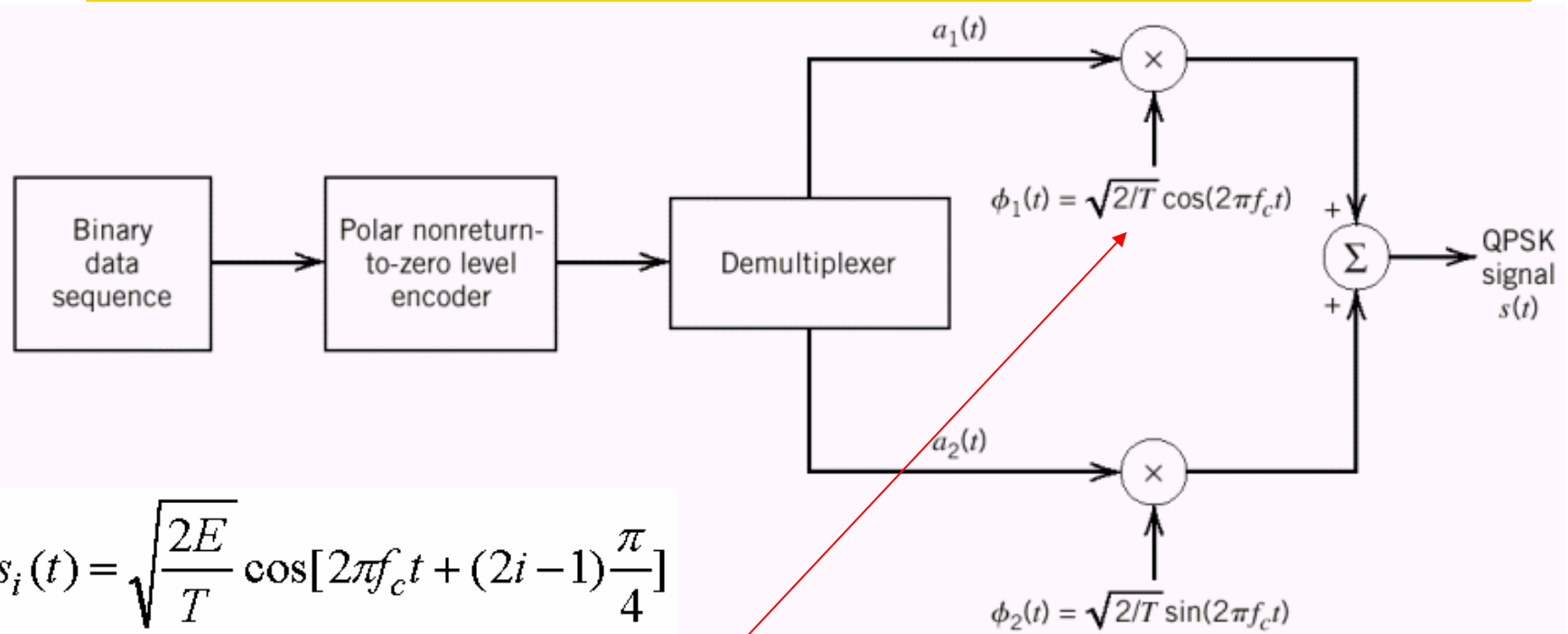
E: Signal energy per *symbol*

T: The symbol duration

Signal Space of QPSK



Generation of QPSK

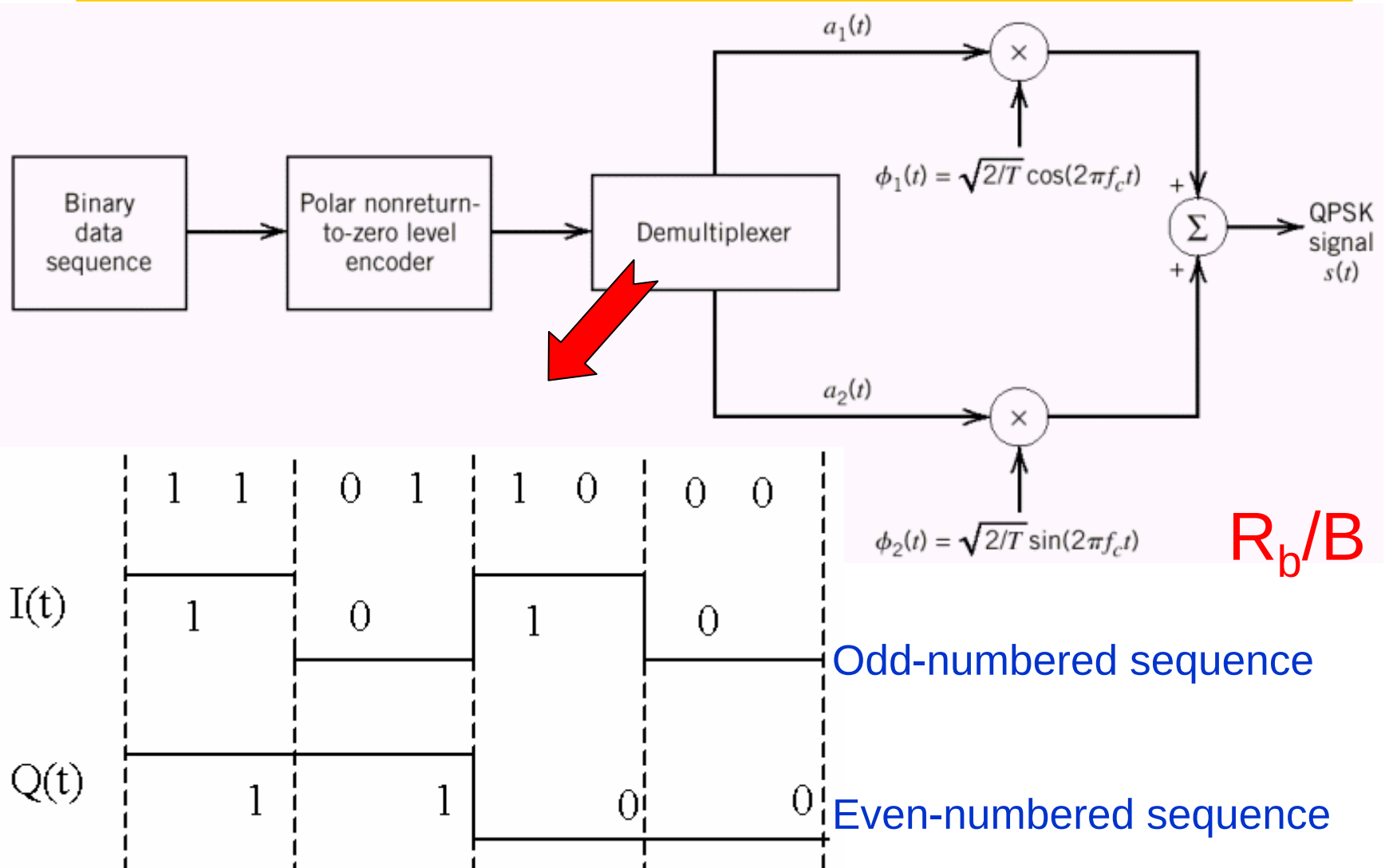


$$s_i(t) = \sqrt{\frac{2E}{T}} \cos\left[2\pi f_c t + (2i-1)\frac{\pi}{4}\right]$$

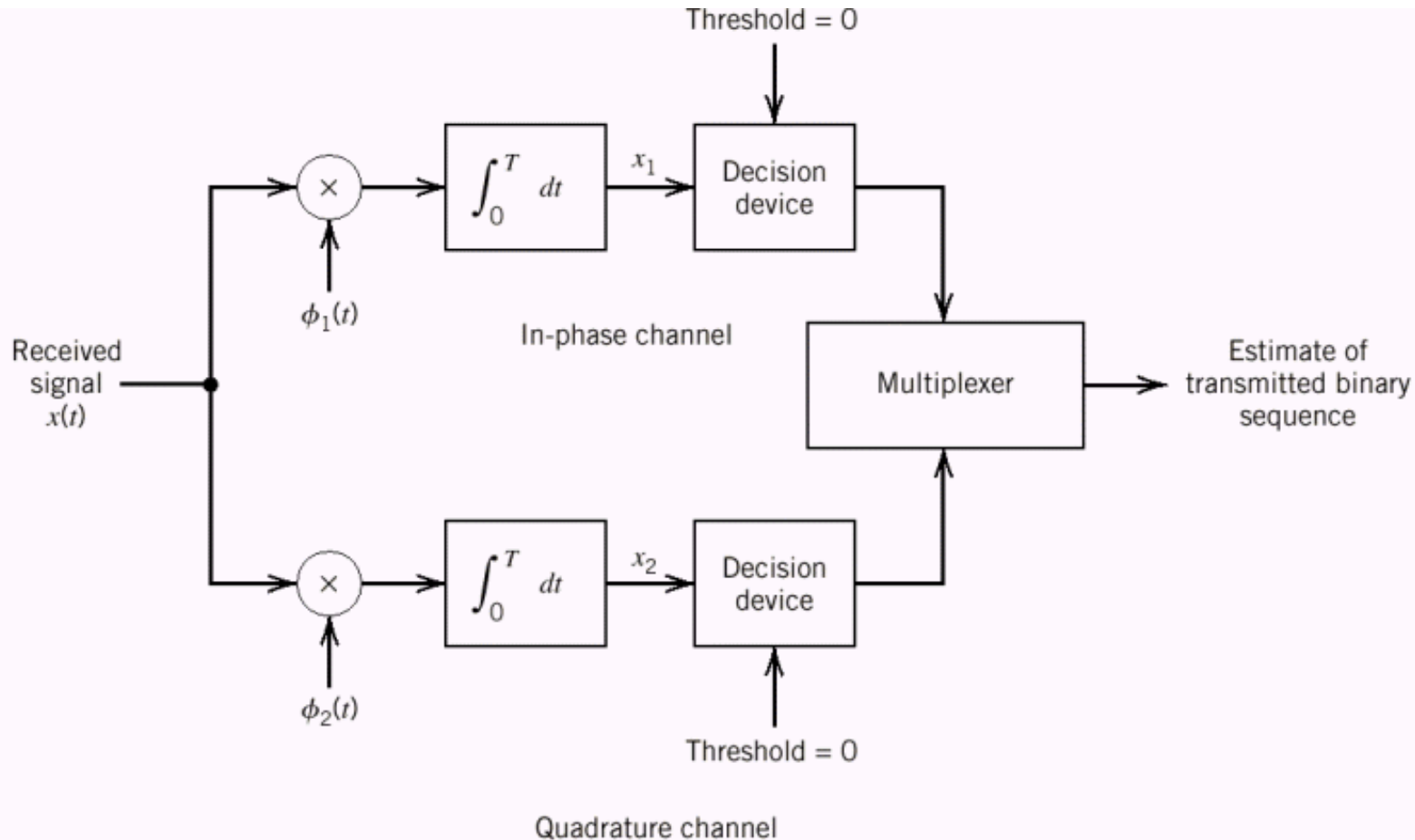
$$= \sqrt{\frac{2E}{T}} \cos\left[(2i-1)\frac{\pi}{4}\right] \cos(2\pi f_c t) - \sqrt{\frac{2E}{T}} \sin\left[(2i-1)\frac{\pi}{4}\right] \sin(2\pi f_c t)$$

$$\text{QPSK} = \text{BPSK}_I + \text{BPSK}_Q$$

Generation of QPSK (2)

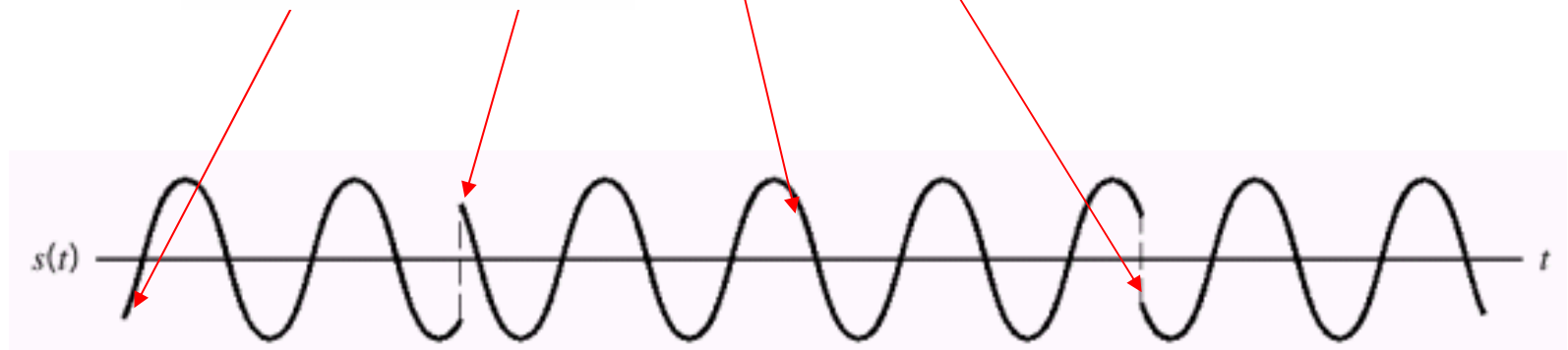
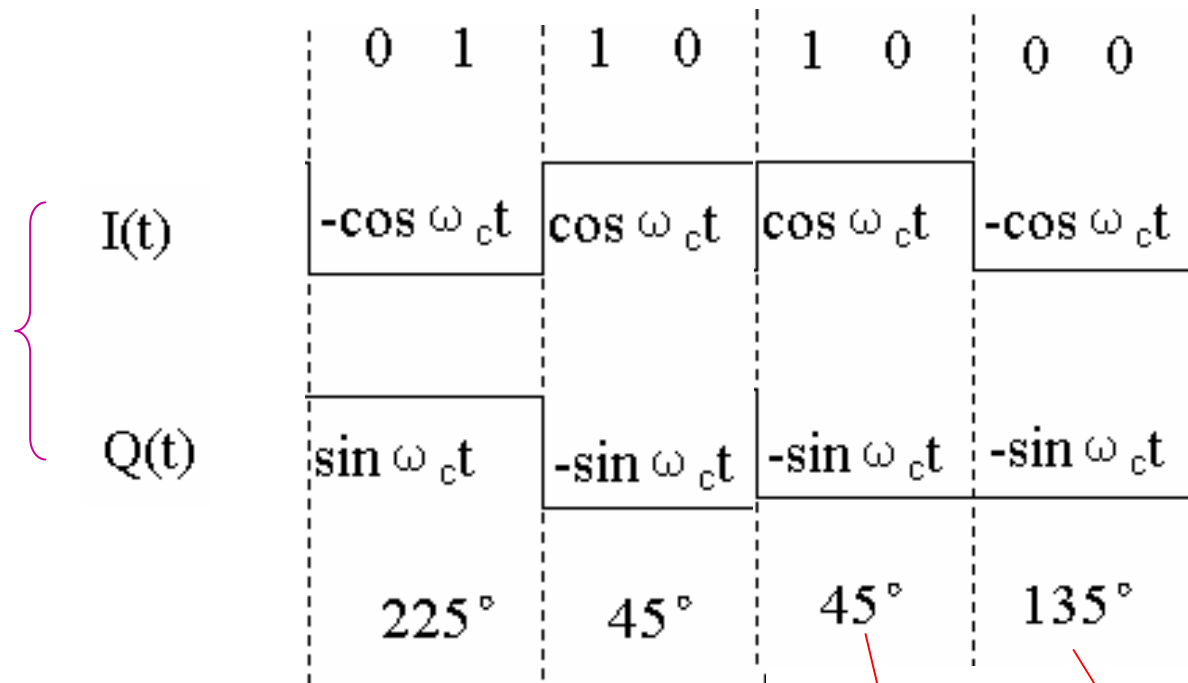


Detection of QPSK



$$\text{QPSK} = \text{BPSK}_I + \text{BPSK}_Q$$

Waveform of QPSK



Review

- **BDPSK** Waveform of “101”
(information code)

Assume: ‘1’ represents changed

‘0’ represents unchanged

P_e of QPSK

- P_e of QPSK using E_b

$$P_e \approx \operatorname{erfc}\left(\sqrt{\frac{E}{2N_0}}\right) = \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

- Bit Error Rate (BER) of QPSK

with Gray Encoding (only 1 bit is changed)

$$BER = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

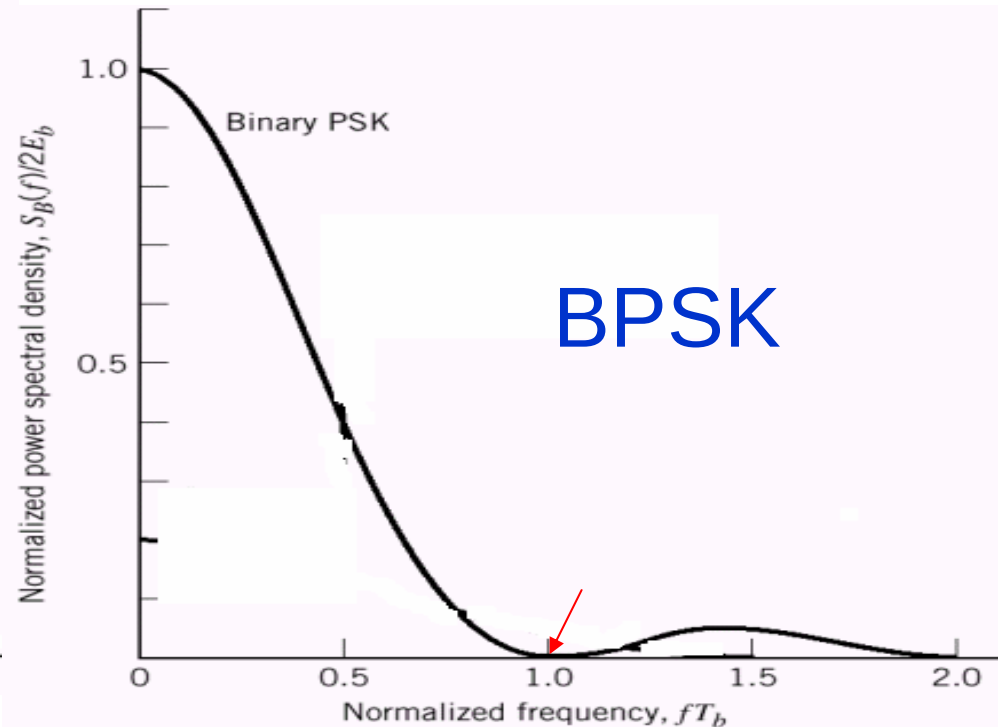
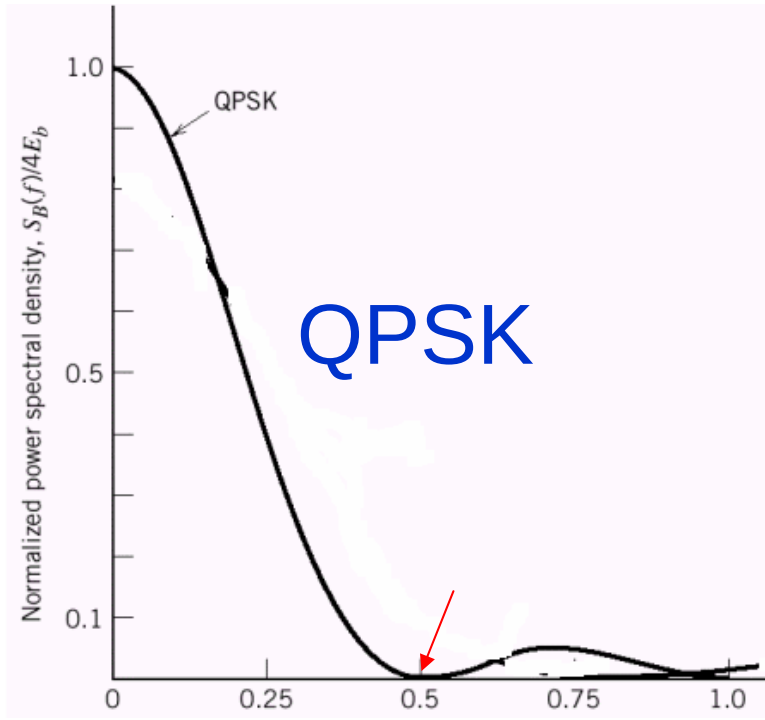
- BER is same as BPSK
- $B \downarrow$ $R_b/B \uparrow$

$$P_{e\text{BPSK}} = p_1 p_{01} + p_0 p_{10} = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

Power Spectra of QPSK

- The baseband power spectral density $S_B(f)$

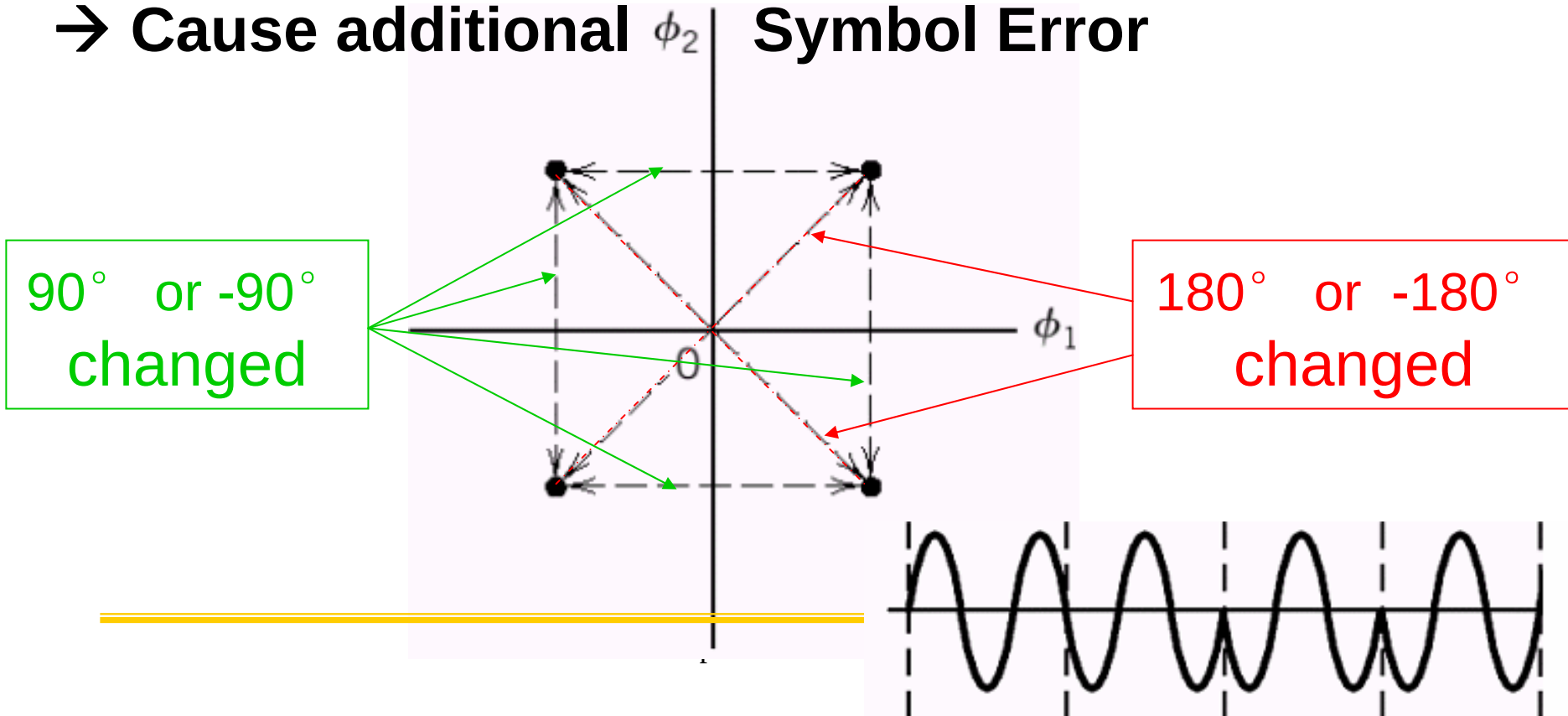
$$S_B(f) = 4E_b \operatorname{sinc}^2(2T_b f)$$



Drawback of QPSK

- 180° and 90° shifts in carrier phase can result in changes in the carrier amplitude (envelope),
Amplitude fluctuation

→ Cause additional ϕ_2 Symbol Error

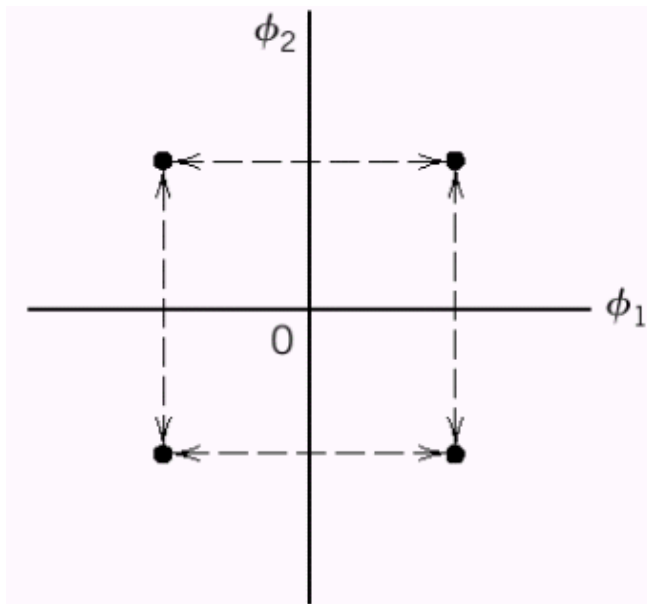


Other QPSK

- Offset QPSK (oQPSK)
- $\pi/4$ -shifted QPSK

Offset QPSK (oQPSK)

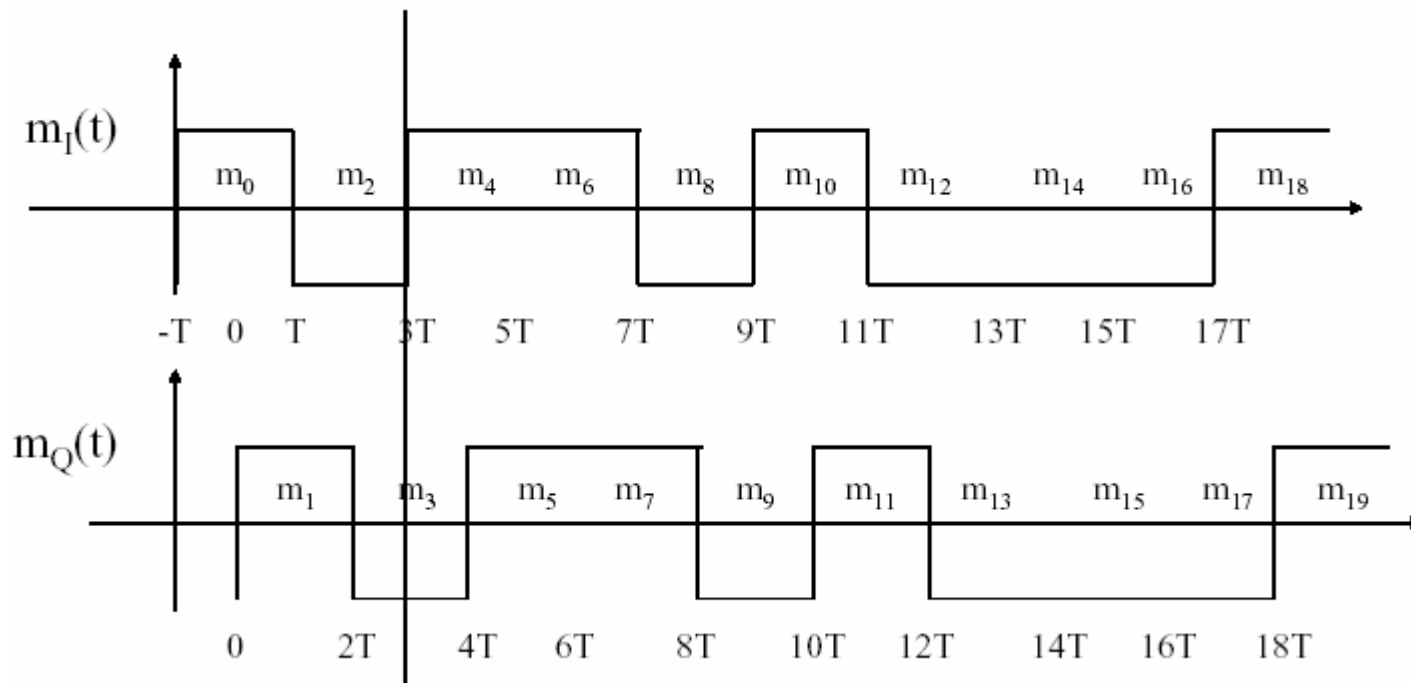
$$\left\{ \begin{array}{ll} \phi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_c t) & 0 \leq t \leq T \\ \phi_2(t) = \sqrt{\frac{2}{T}} \sin(2\pi f_c t) & T/2 \leq t \leq 3T/2 \end{array} \right.$$



- Quadrature component is delayed (i.e. **offset**) by $T/2$.
- **Amplitude fluctuation** ↓

Offset QPSK (oQPSK)

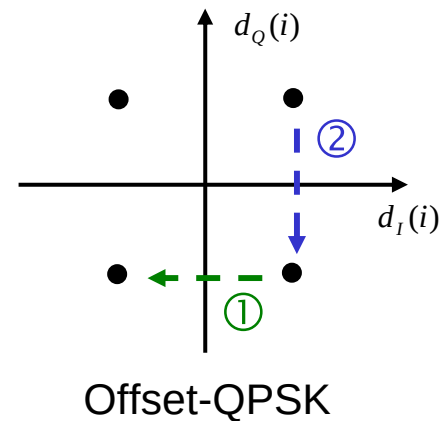
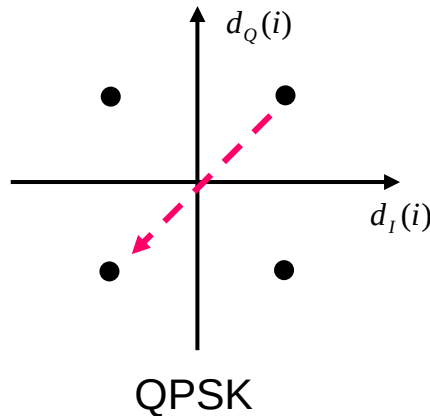
- Phase transitions happen at every half of a symbol period (instead of every symbol period as in QPSK), but each transition is limited to $\pm 90^\circ$



Offset QPSK (oQPSK)

$$s_{oQPSK}(t) = \sum_{i=-\infty}^{\infty} d_I(i) \cdot g_{Tx}(t - iT) + j \cdot \sum_{i=-\infty}^{\infty} d_Q(i) \cdot g_{Tx}(t - \frac{T}{2} - iT)$$

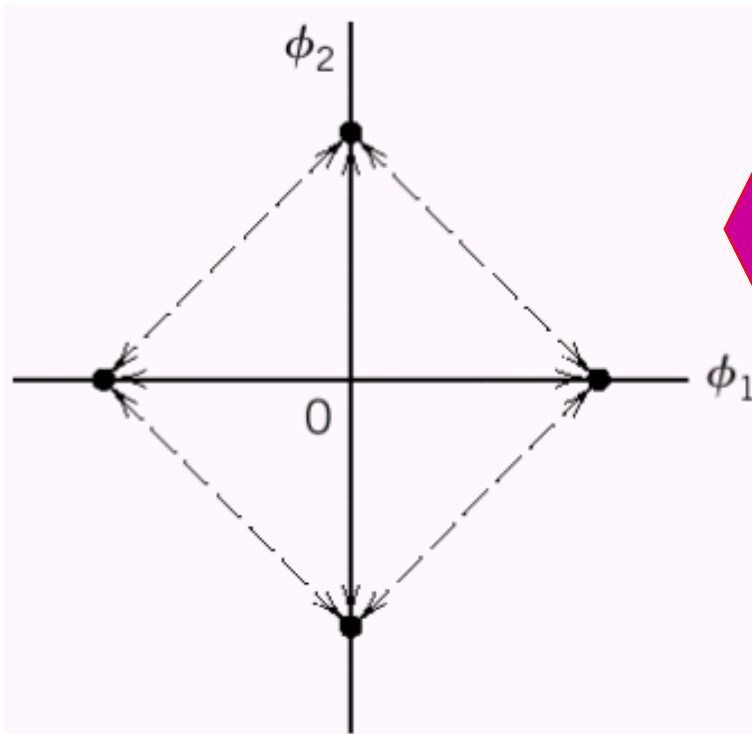
- ⇒ symbol transitions in **two steps**:
- first in-phase component
 - **T/2 later** quadrature component



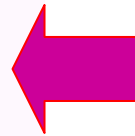
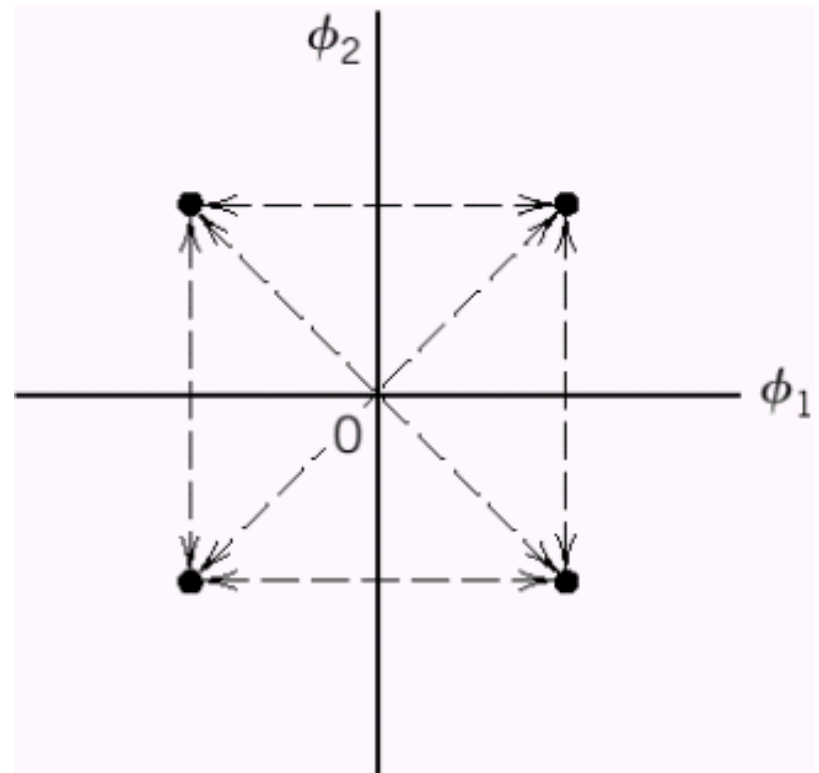
⇒ no zero crossing

$\pi/4$ -shifted QPSK

Constellation of
 $\pi/4$ -shifted QPSK



Constellation of
QPSK



$\pi/4$ -shifted QPSK

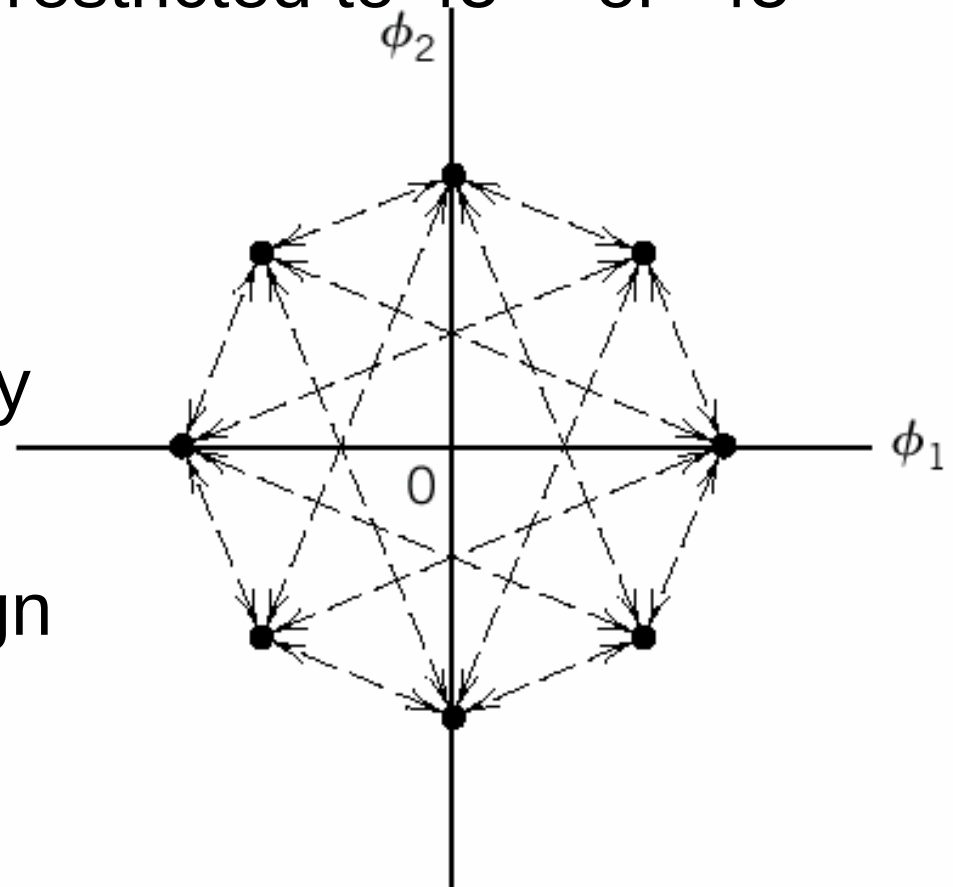
- Advantages:

- Phase transitions are restricted to 45° or -45° or 135° or -135°

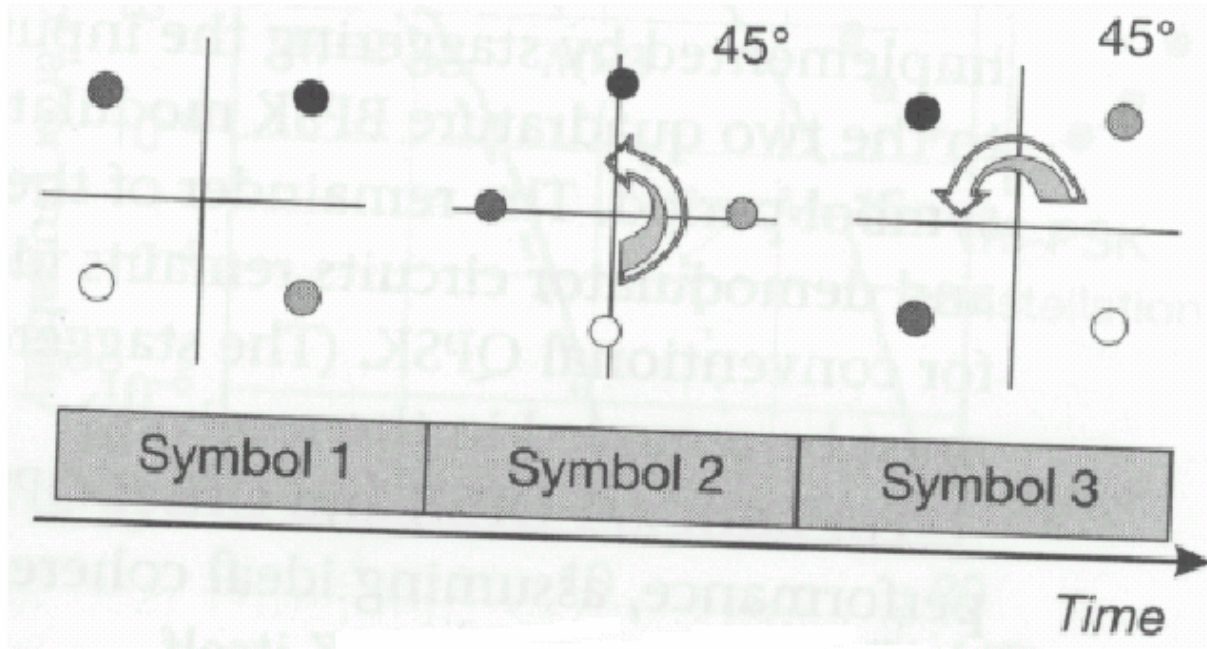
→ envelope variation ↓

- Can be non-coherently detected

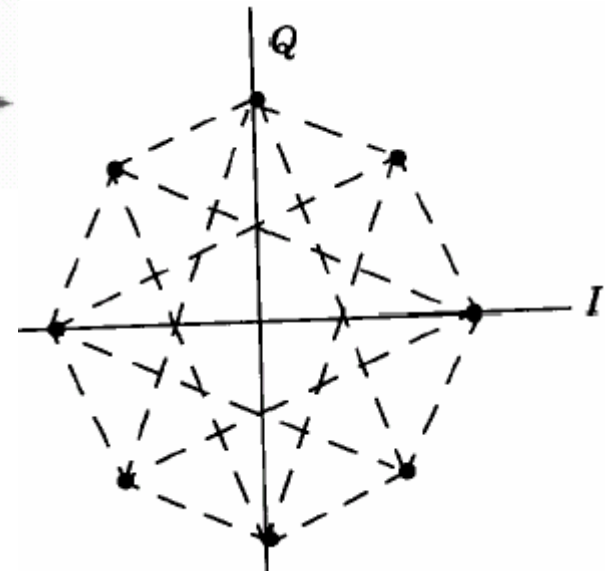
→ Simply receiver design



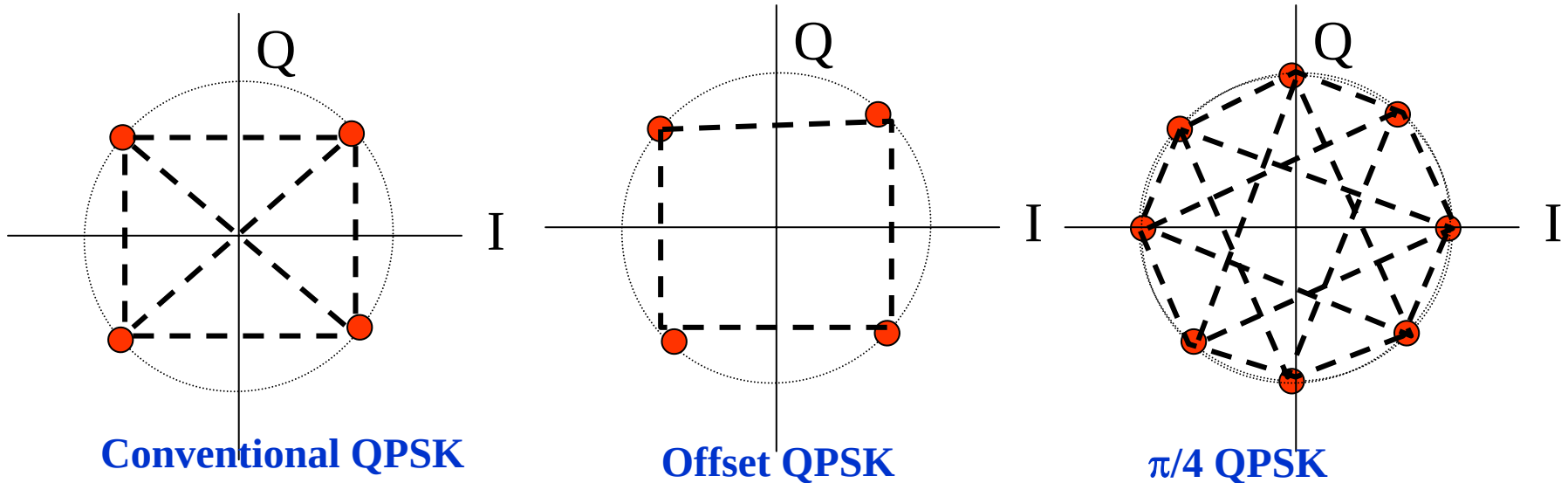
$\pi/4$ -shifted QPSK



All possible phase change:



Types of QPSK



- **Conventional QPSK** has transitions through zero (i.e. 180° phase transition). Highly linear amplifiers required.
- In **Offset QPSK**, the phase transitions are limited to 90° , the transitions on the I and Q channels are staggered.
- In **$\pi/4$ QPSK** the set of constellation points are toggled each symbol, so transitions through zero cannot occur. This scheme produces the **lowest** envelope variations.

MPSK

M-ary PSK

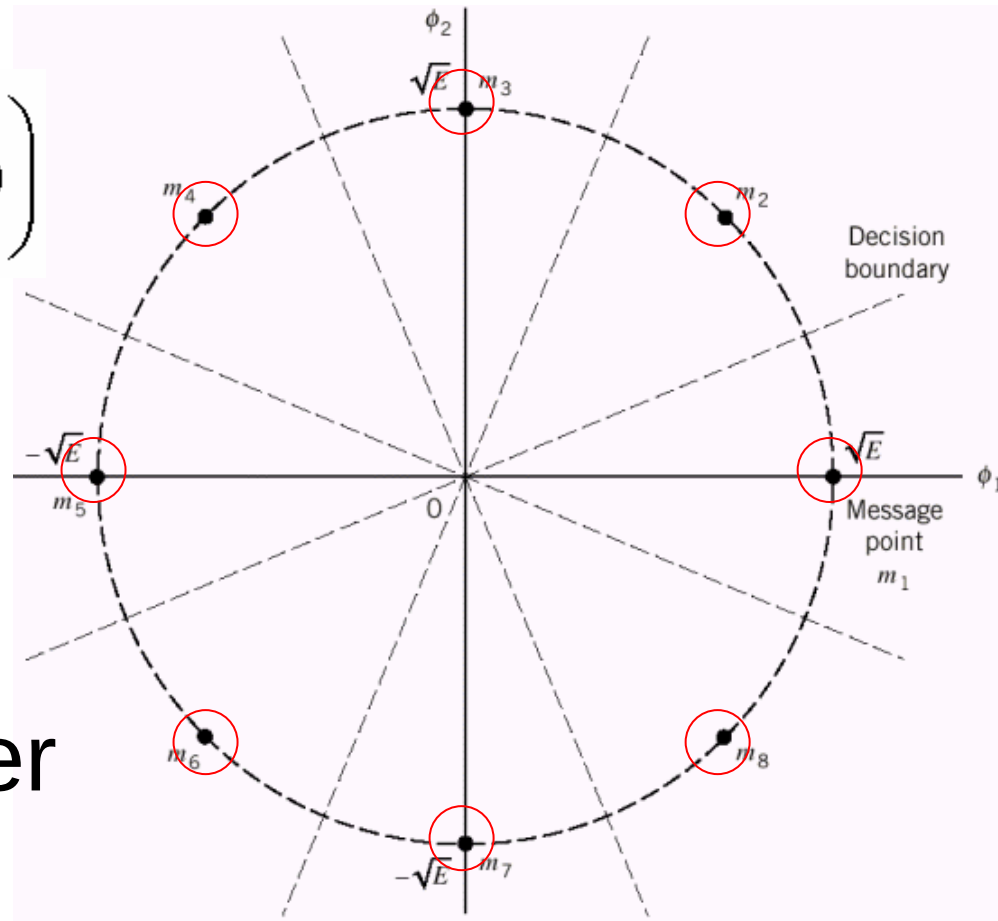
- M=8 **Octa**-Phase Shift Keying

$$s_i = \sqrt{\frac{2E}{T}} \cos\left(\omega_c t + \frac{2\pi}{M}(i-1)\right)$$

$$i=1,2,\dots,M$$

E: symbol energy

$f_c = n_c/T$, n_c : integer



P_e of 8PSK

$$P_e \leq \frac{1}{2} \sum_{k=1, k \neq i}^M \operatorname{erfc}\left(\frac{d_{ik}}{2\sqrt{N_0}}\right)$$

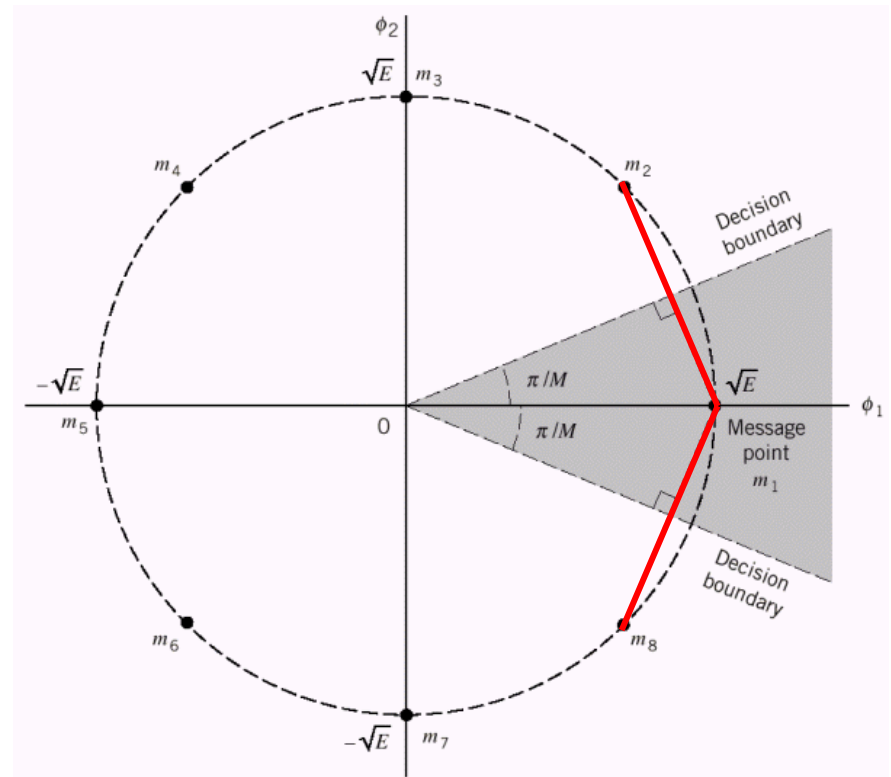
- E/N_0 is large enough to consider the nearest two message points

$$d_{12} = d_{18} = 2\sqrt{E} \sin\left(\frac{\pi}{M}\right)$$

$$P_e \approx \operatorname{erfc}\left(\sqrt{\frac{E}{N_0}} \sin\left(\frac{\pi}{M}\right)\right)$$

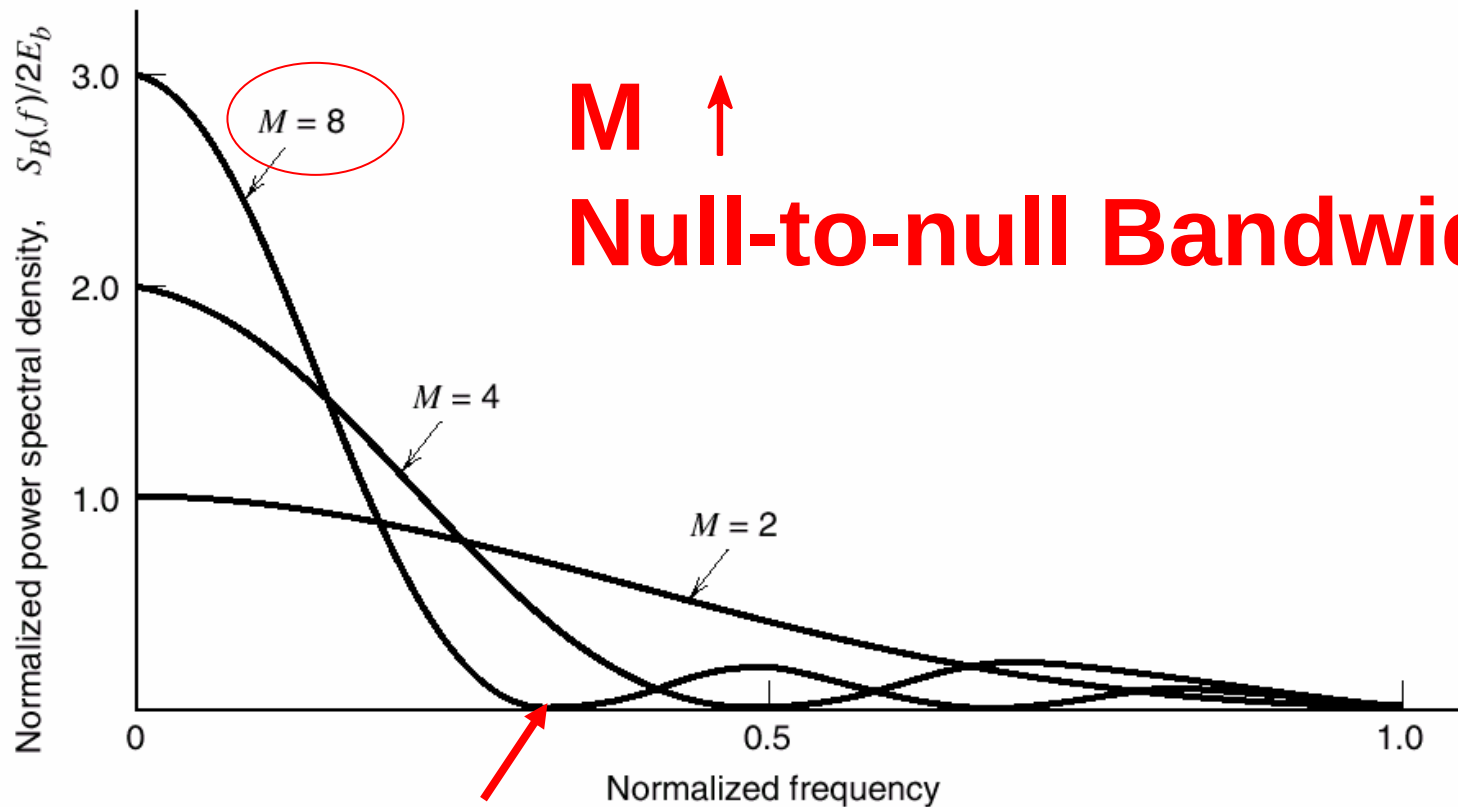
Assumed $M \geq 4$

M ↑ P_e ↑



Power Spectra of MPSK

$$S_B(f) = 2E_b \log_2 M \operatorname{sinc}^2(T_b f \log_2 M)$$



Spectral nulls: frequencies at which psd is zero

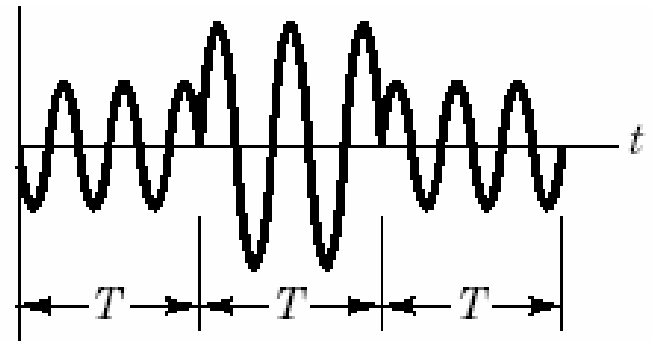
Power Spectra of MPSK (2)

- As M is increased, the **bandwidth efficiency** is improved at the expense of **error performance**.
- To ensure no degradation in error performance, we have to increase E_b/N_0 to compensate for the increase in M .

APK(ASK/PSK)

ASK/PSK (APK)

$$s_i(t) = \sqrt{\frac{2E_i(t)}{T}} \cos [\omega_0 t + \phi_i(t)]$$
$$i = 1, 2, \dots, M$$
$$0 \leq t \leq T$$



APK

- **M-ary PSK**

in-phase and quadrature components
are interrelated

- **M-ary QAM:**

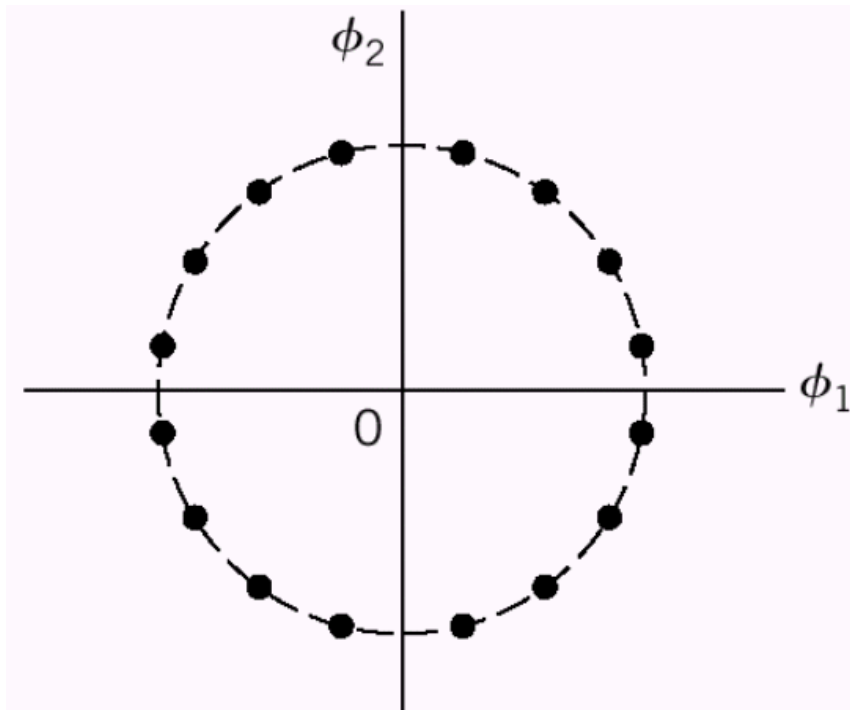
M-ary Quadrature Amplitude Modulation

in-phase and quadrature components
are independent

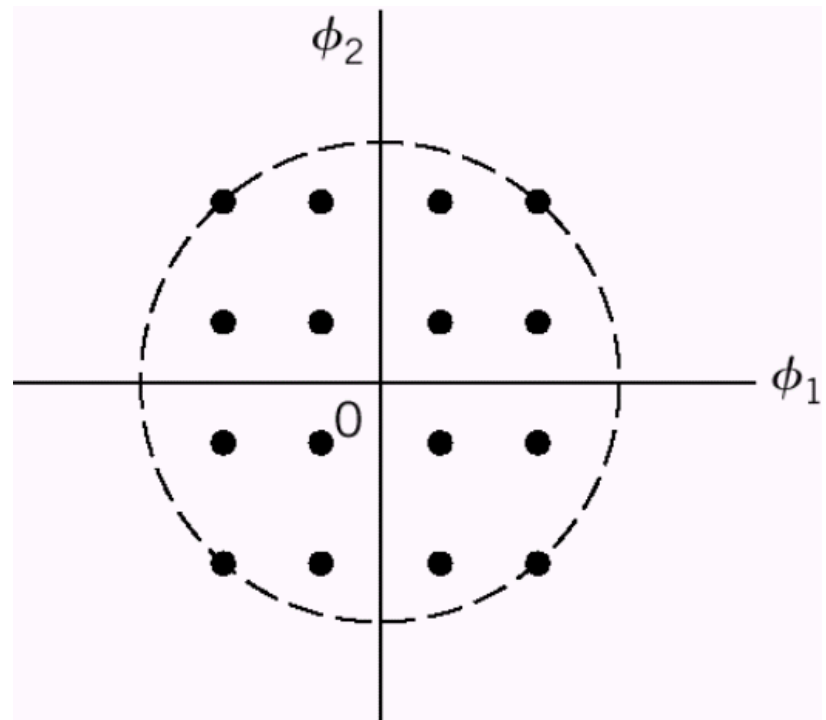
MPSK versus MQAM

- e.g. $M=16$

16PSK

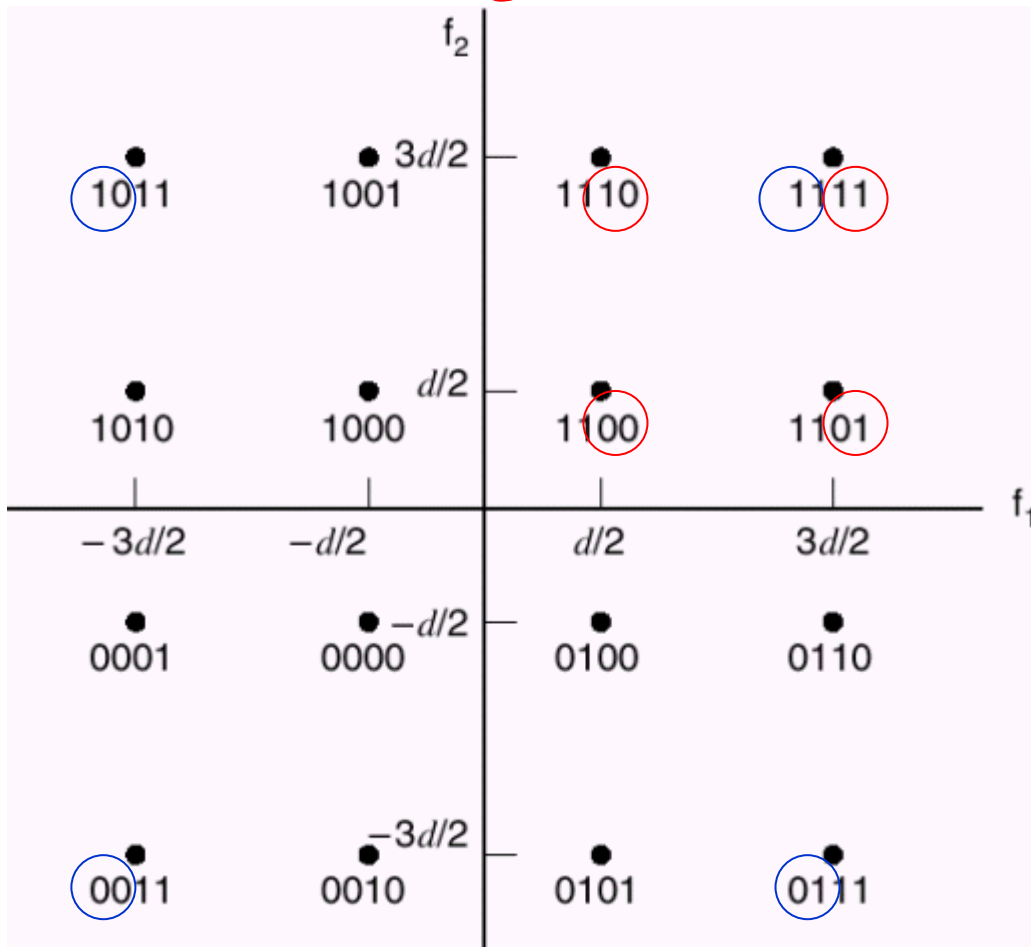


16QAM



MQAM Square Constellations

- **16QAM**



- Quadbits

- left-most 2 bits:

$10 \leftarrow 11$

$00 \rightarrow 01$

- remaining 2 bits:

$10 \leftarrow 11$

$00 \rightarrow 01$

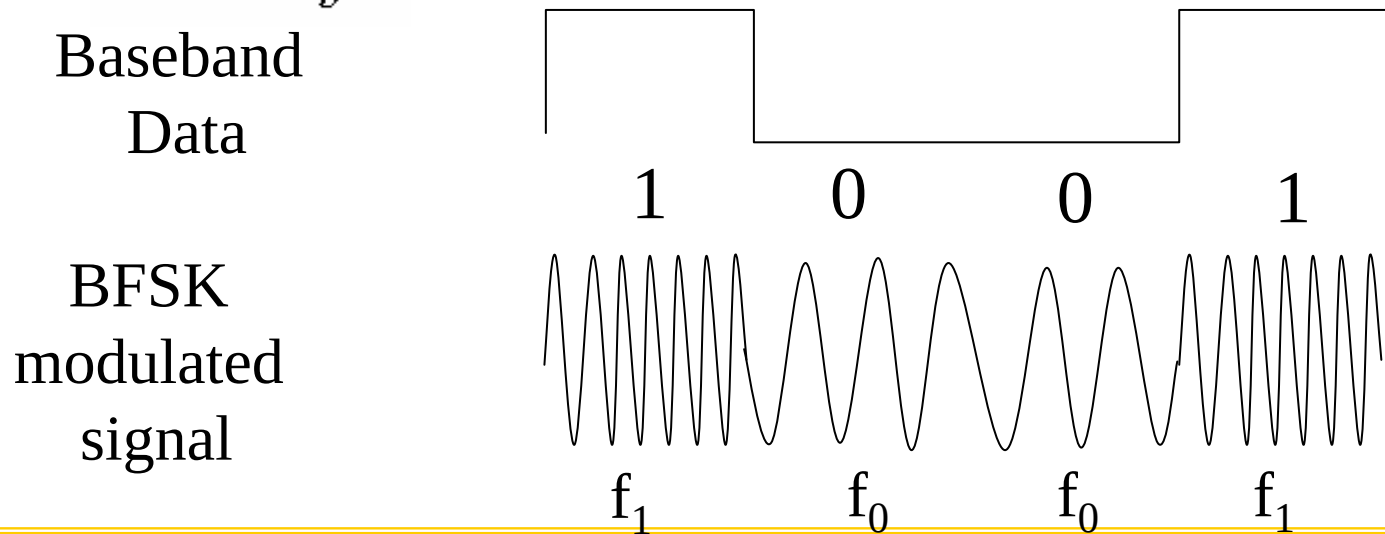
FSK

BFSK

- Binary Frequency Shifted- Keying (BFSK)

Definition: $s_i(t) = \sqrt{\frac{2E_b}{T_b}} \cos[2\pi f_i t] \quad 0 \leq t \leq T_b$

$$f_i = \frac{n_c + i}{T_b} \quad \text{For some fixed integer } n_c, i=1,2$$

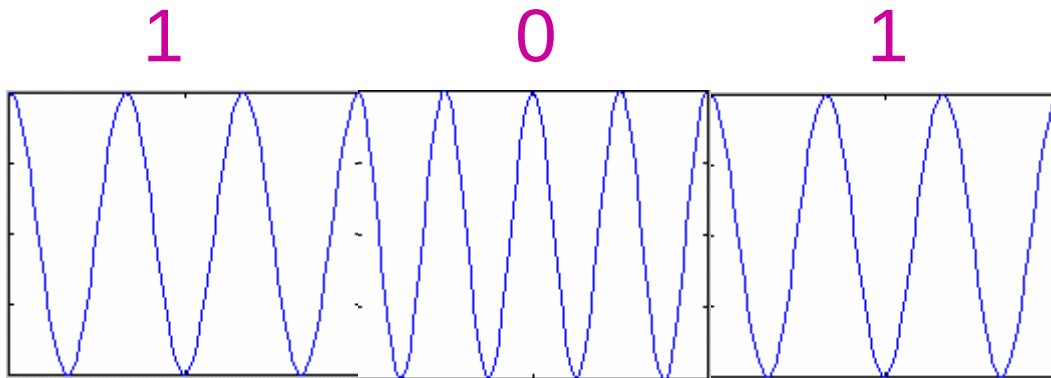


BFSK (2)

$$f_i = \frac{n_c + i}{T_b}$$

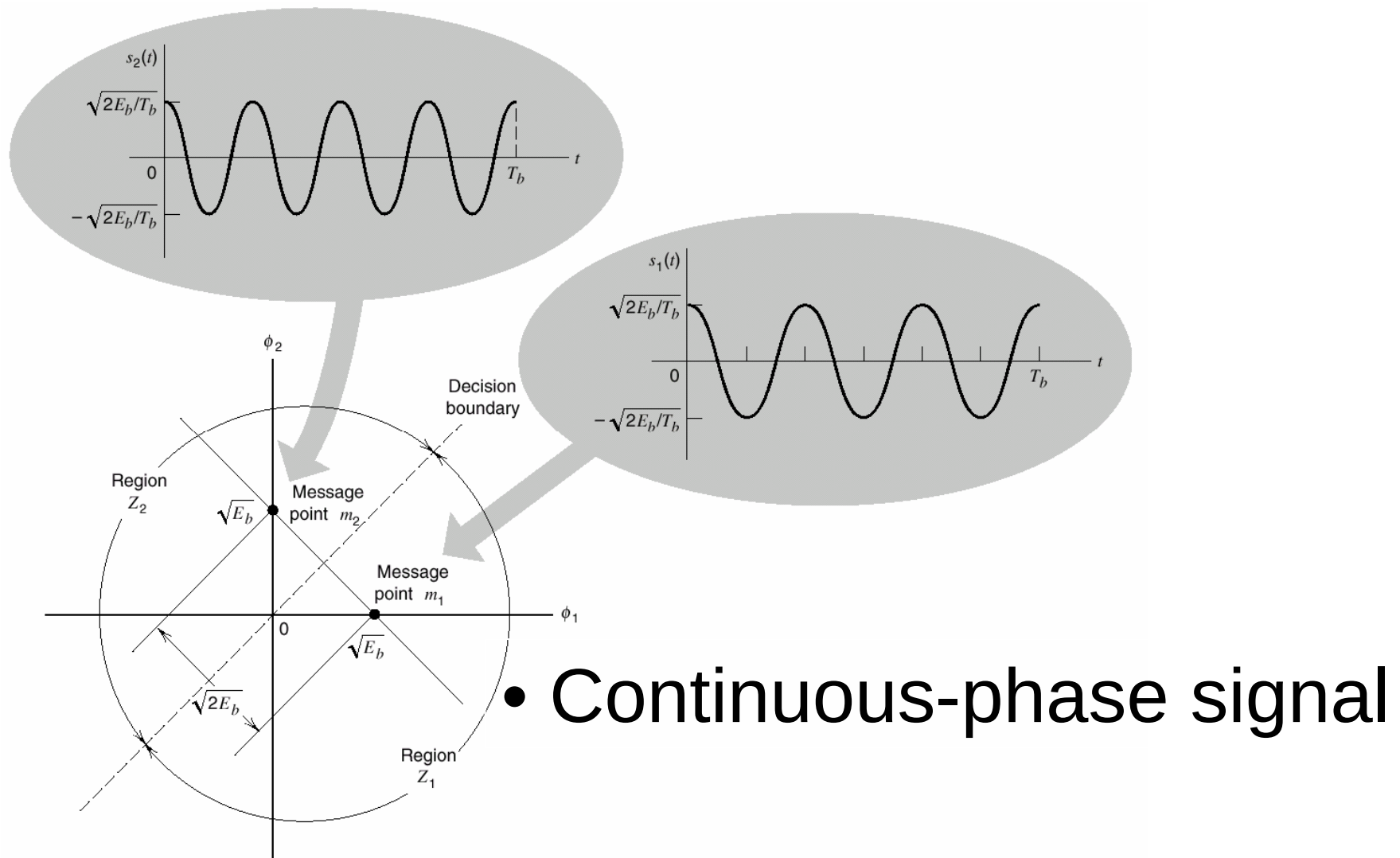
- e.g. $f_1 = 3/T_b$ $s_1(t)$ “1”
 $f_2 = 4/T_b$ $s_2(t)$ “0”

• Sunde's FSK



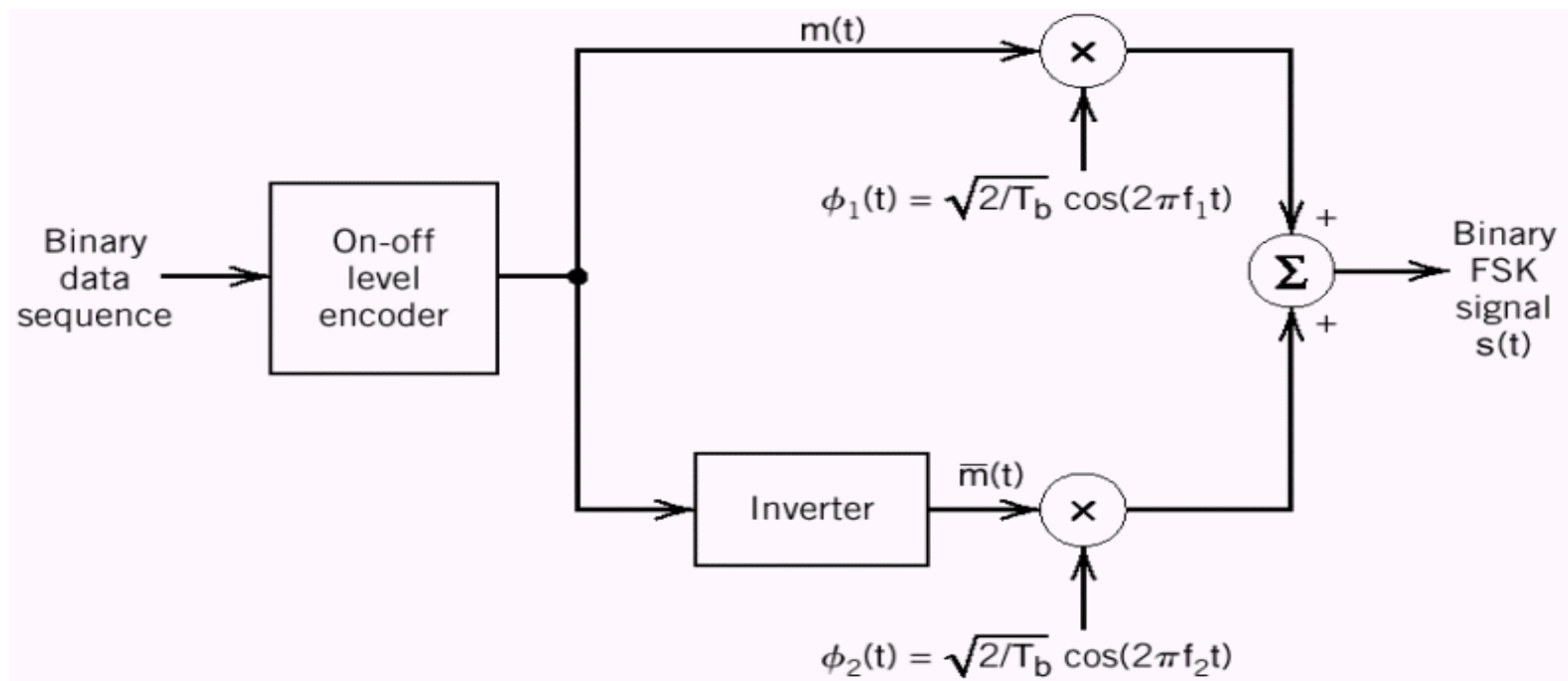
- **Continuous-phase FSK (CPFSK):**
Phase continuity is always maintained

Signal Space of BFSK



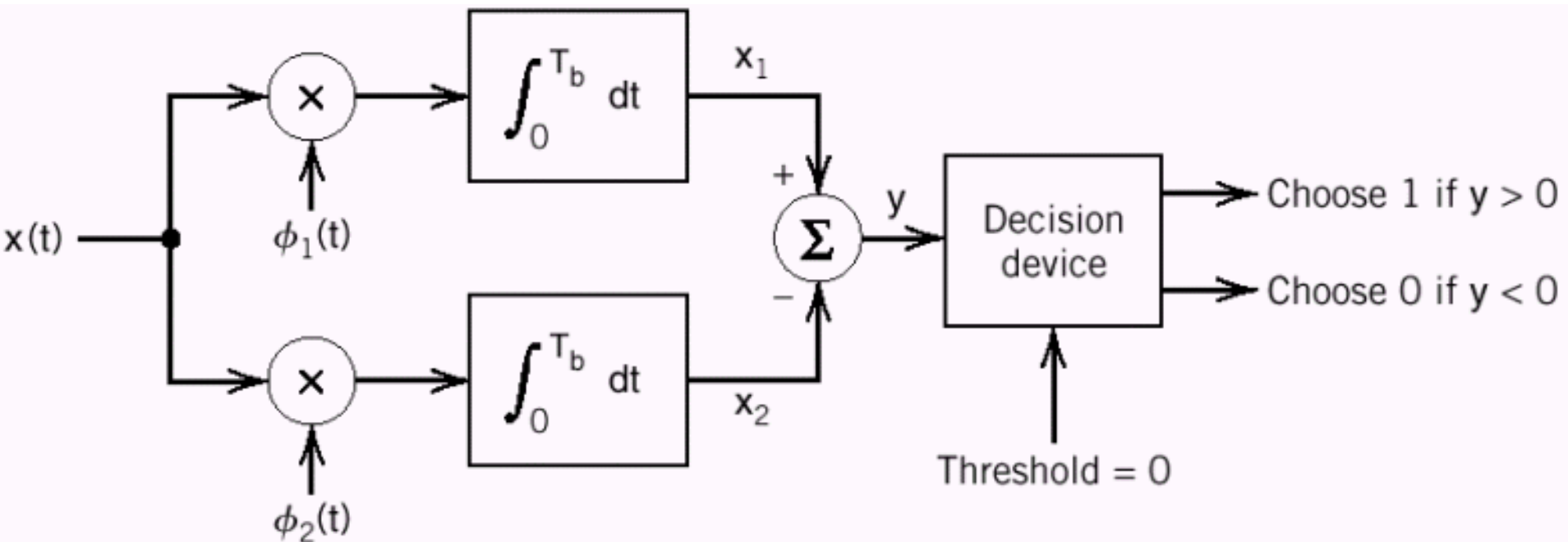
Generation of BFSK

- On-off level encoder: $\begin{cases} \text{"1"} & \sqrt{E_b} \text{ volts} \\ \text{"0"} & \text{Zero volt} \end{cases}$
- Inverter: $\rightarrow f_1$ or f_2



Detection of BFSK

- f_1 or f_2 , at any time



P_e of BFSK

- **BFSK:**

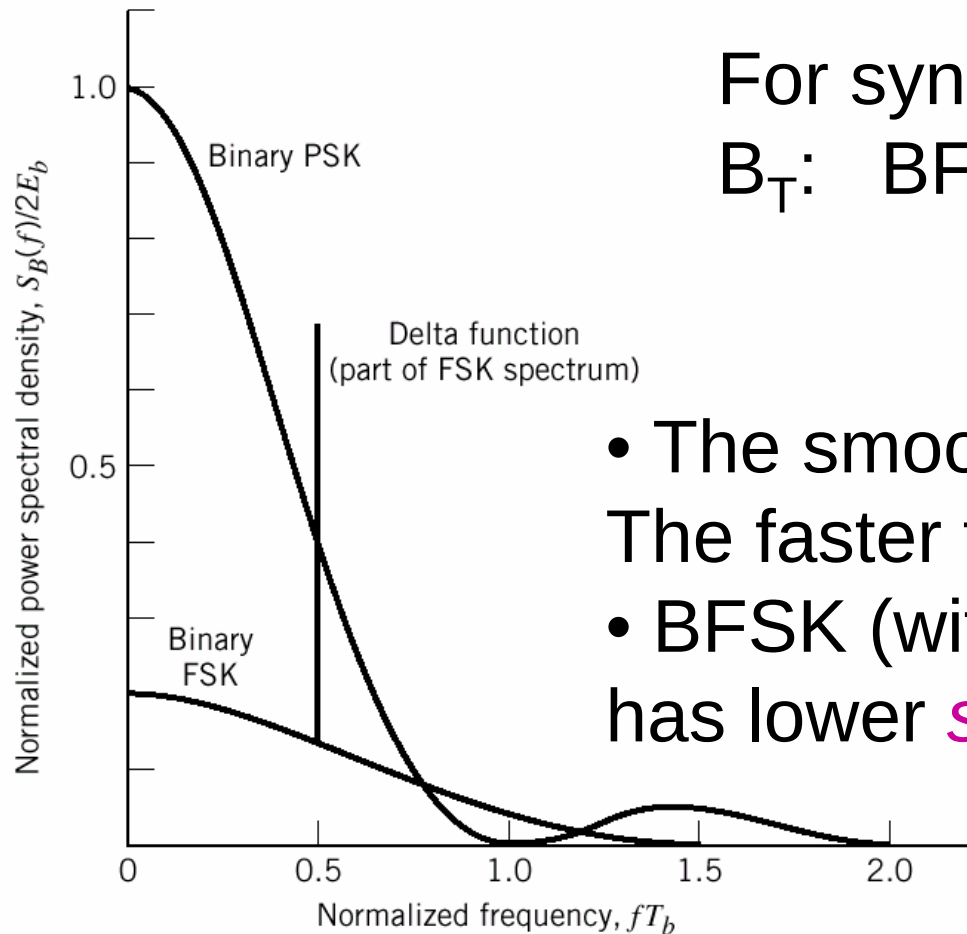
$$P_e = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{2N_0}}\right)$$

- **BPSK:**

$$P_e = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

- P_e performance: **BPSK > BFSK**
- In perfect accord with signal space, i.e. $d_{\min}$ of BPSK is larger than BFSK's

Power Spectra of BFSK



For synchronization
 B_T : BFSK > BPSK

- The smoother the pulse,
The faster the drop of tail to zero
- BFSK (with continuous phase)
has lower *side-lobes*

→ has *discrete frequency* components $1/2T_b$

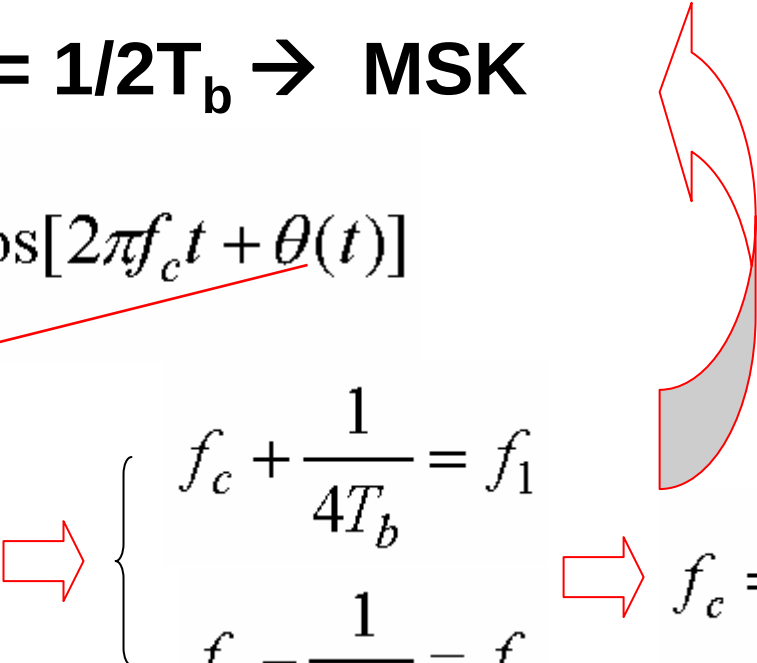
MSK

- Minimum Shift Keying (MSK)

CPFSK $\rightarrow f_2 - f_1 = 1/T_b$ Deviation ratio: $h = 1/2$

$f_2 - f_1 = 1/2T_b \rightarrow$ MSK

CPFSK: $s(t) = \sqrt{\frac{2E_b}{T_b}} \cos[2\pi f_c t + \theta(t)]$

$$\theta(t) = \theta(0) \pm \frac{\pi t}{2T_b} \Rightarrow \begin{cases} f_c + \frac{1}{4T_b} = f_1 \\ f_c - \frac{1}{4T_b} = f_2 \end{cases} \Rightarrow f_c = \frac{1}{2}(f_1 + f_2)$$


MSK (2)

- At time $t = T_b$:

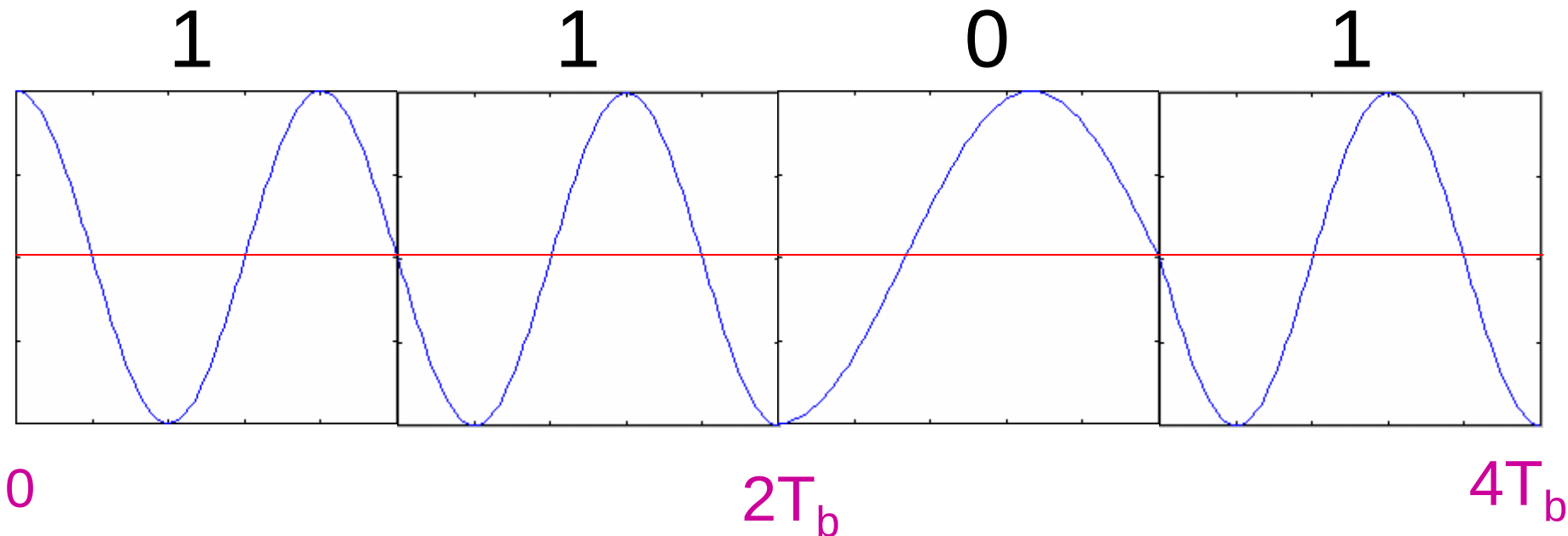
$$\theta(t) = \theta(0) \pm \frac{\pi t}{2T_b} \Rightarrow \theta(T_b) - \theta(0) = \begin{cases} \pi/2 & \text{"1"} \\ -\pi/2 & \text{"0"} \end{cases}$$

- sending "1": **increase** phase of CPFSK signal $s(t)$ by $\pi/2$;
- sending "0": **decrease** phase of CPFSK signal $s(t)$ by $\pi/2$;

Waveform of MSK

- e.g. sequence : 1 1 0 1

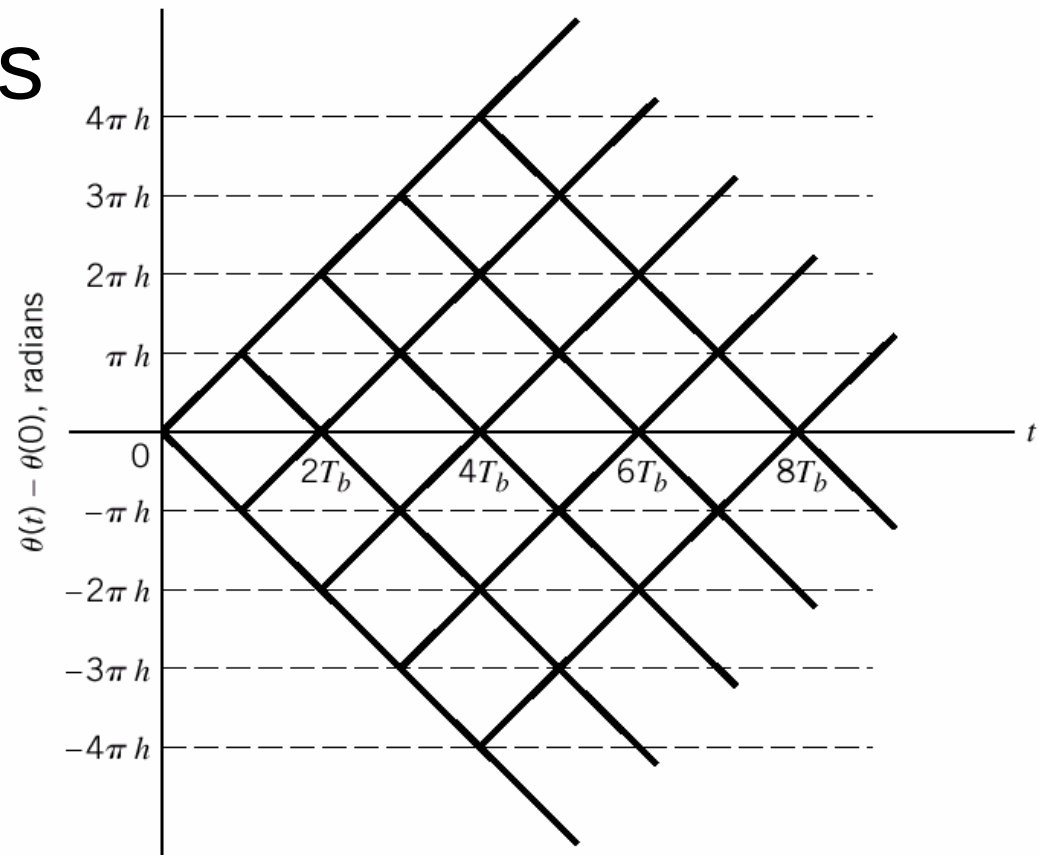
$$\begin{cases} f_1 = 5/4T_b \\ f_2 = 3/4T_b \end{cases} \quad f_c = 1/T_b$$



Phase Tree

All possible paths

$$\theta(t) - \theta(0) = \pm \frac{\pi h}{T_b} t$$

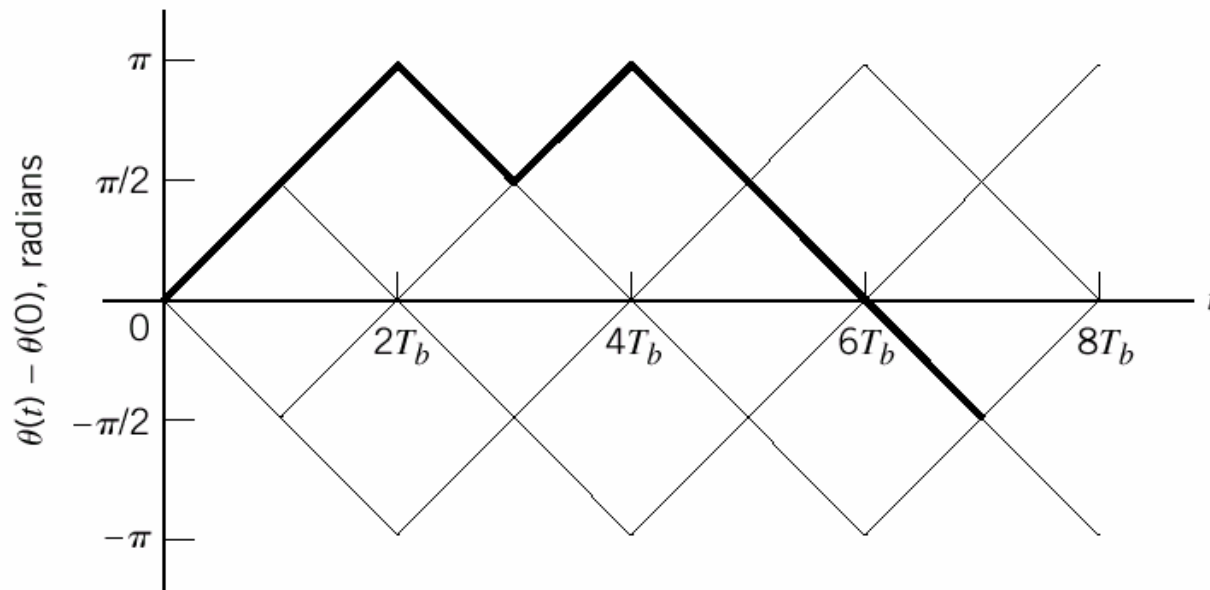


- A manifestation of phase continuity

Phase Trelis

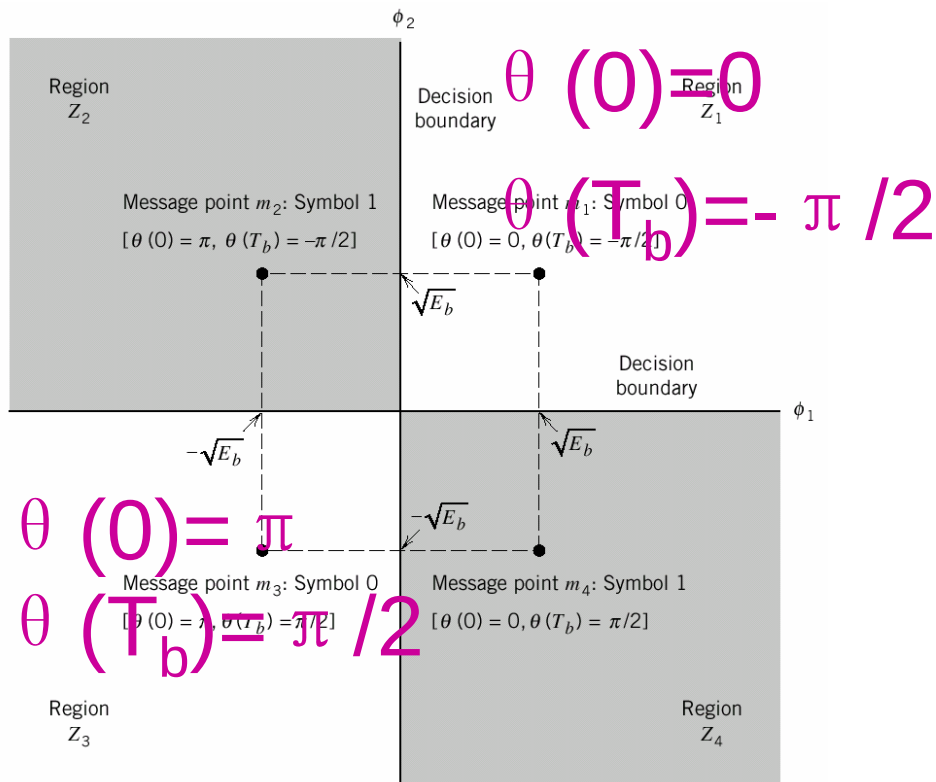
- when deviation ratio $h = 1/2$

1 1 0 1 0 0 0 with $\theta(0)=0$

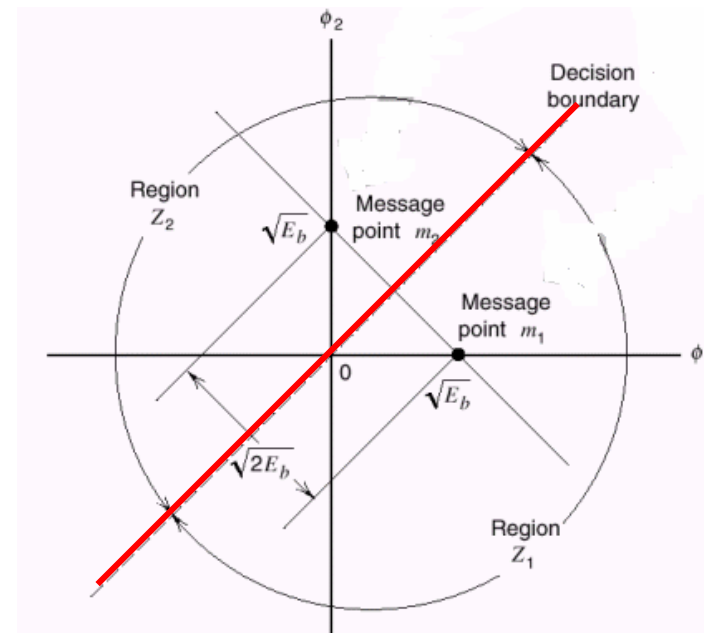


Signal Space of MSK

• MSK

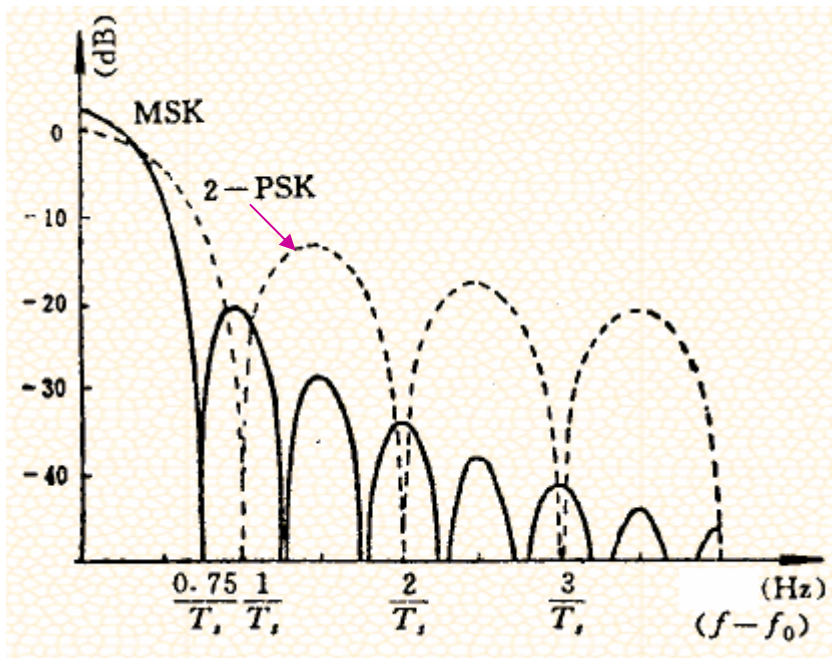


• BFSK

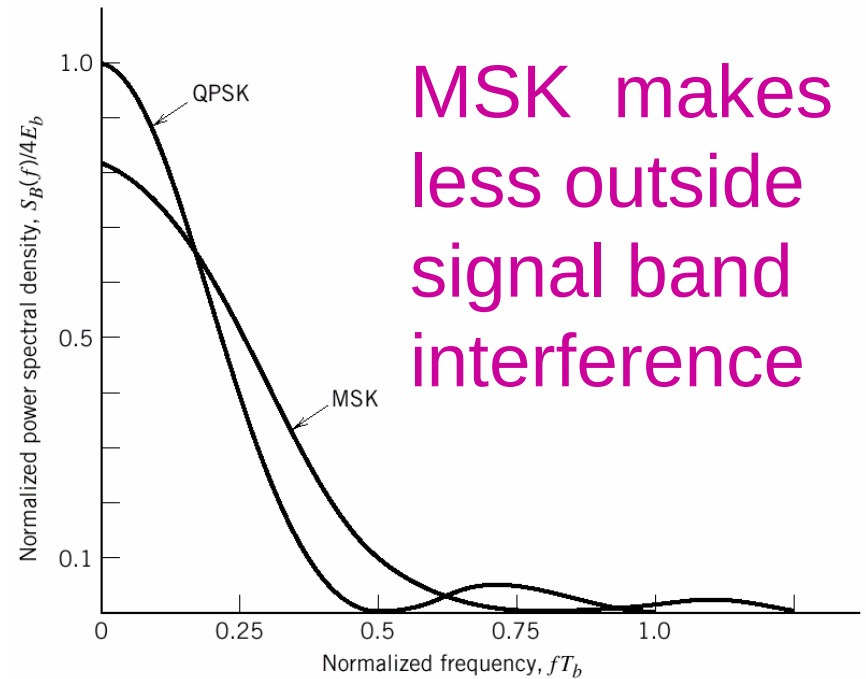


Power Spectra of MSK

- Minimum Shift Keying (MSK)



B_T : MSK < BPSK



MSK makes less outside signal band interference

MSK :	f	↑	Tail	↓	↓	↓
QPSK:	f	↑	Tail	↓	↓	

P_e of MSK

- MSK:

$$P_e = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

- BPSK:

$$P_e = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

- P_e performance: **BPSK = MSK**
- On the basis of observation over $2T_b$

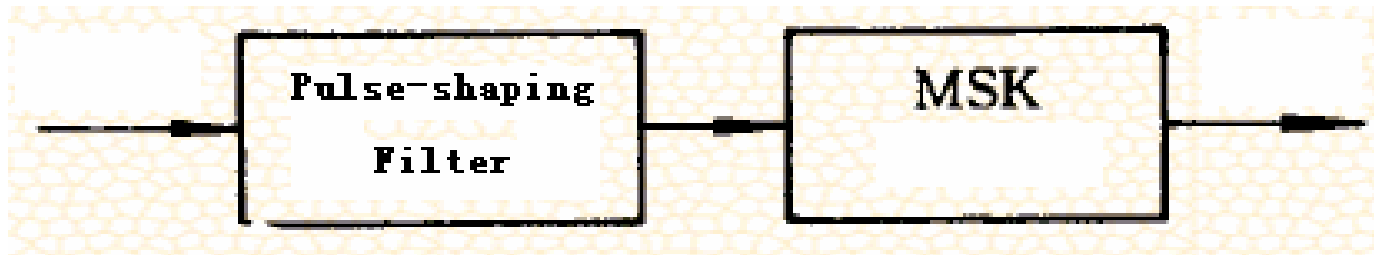
Summary of MSK

- Constant envelope
- Relatively narrow bandwidth
- Coherent detection performance equivalent to that of QPSK
- Disadvantage:

MSK does not satisfy the strict requirement in wireless communication

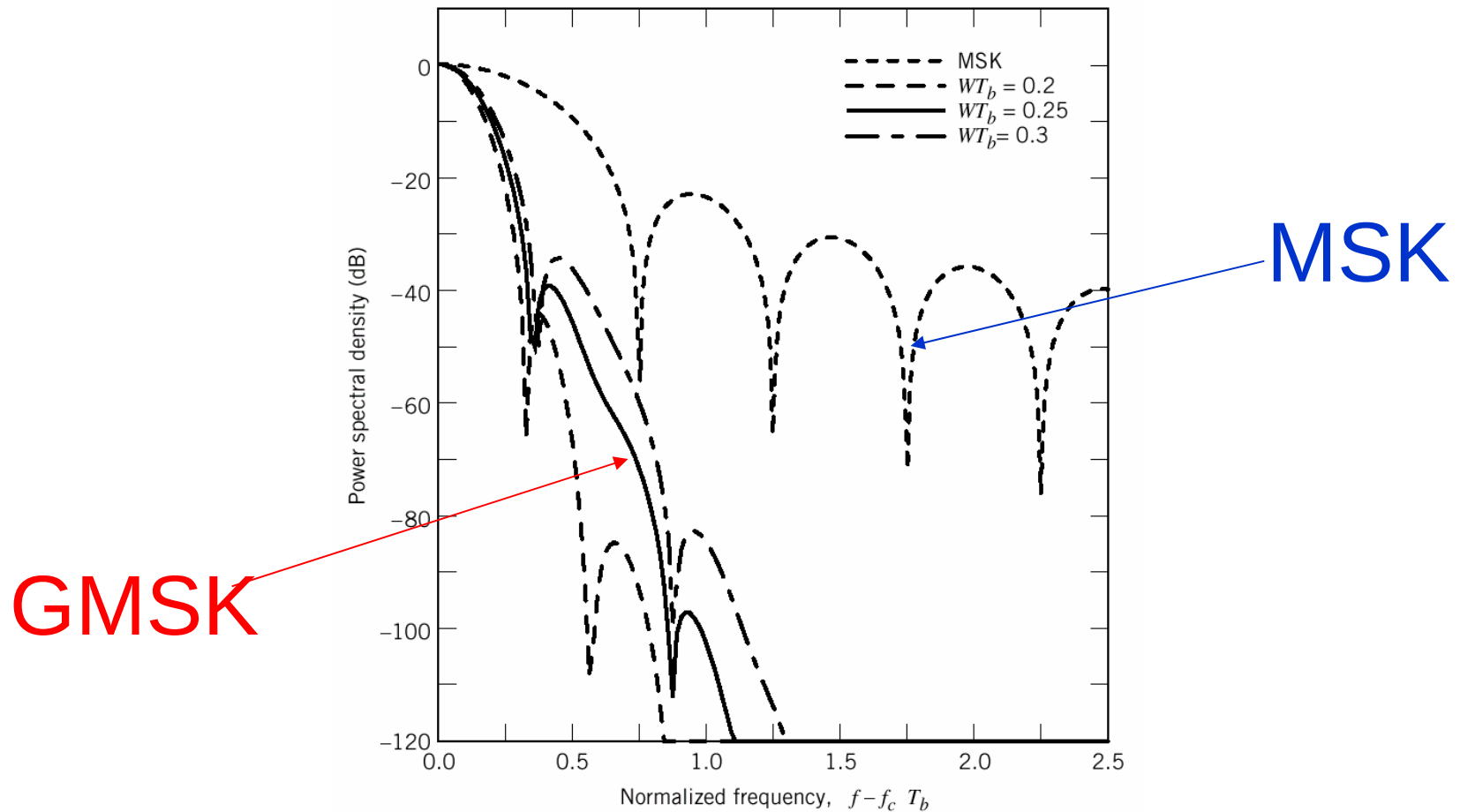
GMSK

- Gaussian-Filtered MSK (GMSK)



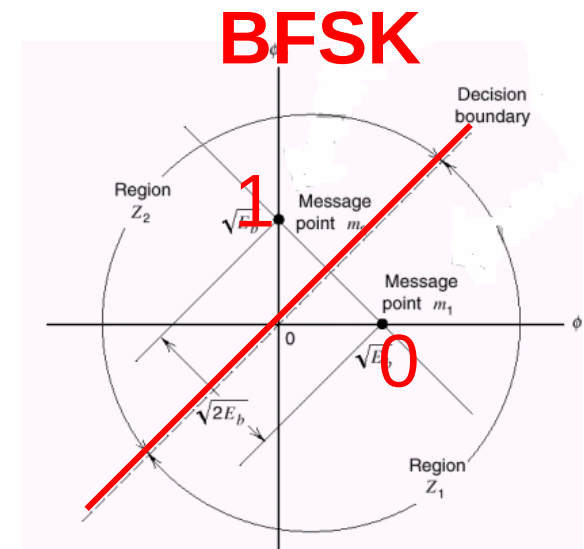
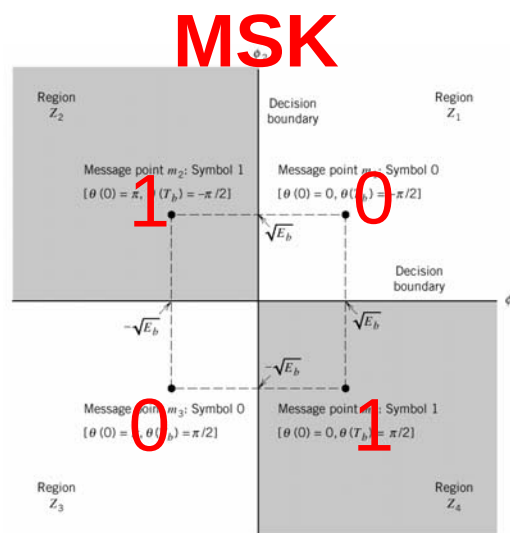
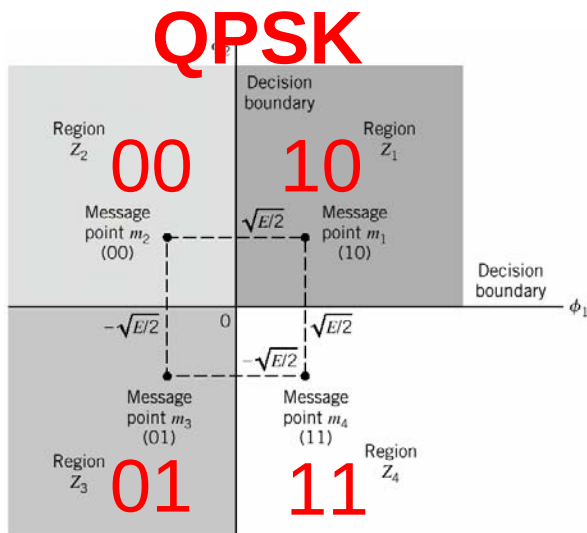
- Pre-modulation LPF:
Frequency with narrow bandwidth
and sharp cutoff characteristics

Power Spectral of GMSK



Comparison of Digital Modulation

- Monotonically **decreasing** function of E_b/N_0
- BPSK, QPSK, MSK produce a smaller BER
- MSK processes 2 successive bit intervals ($2T_b$) to recover original information code $\rightarrow$ has **memory**

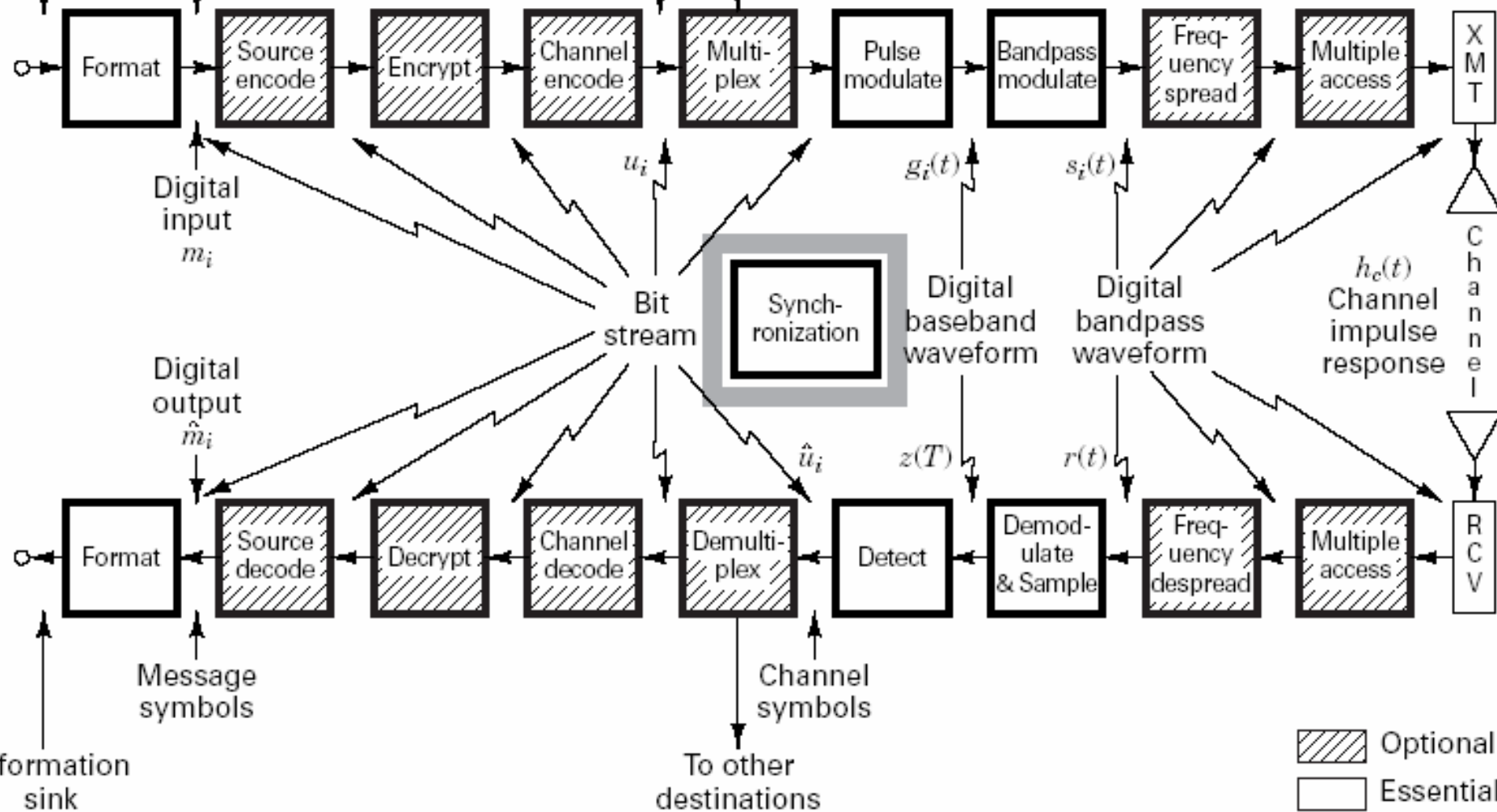


Information source

Message symbols

From other sources

Channel symbols



Synchronization

- 
- **Carrier synchronization**
 - **Symbol synchronization**
 - **Frame synchronization**

E.g. ...

Synchronization

- Data-aided synchronization

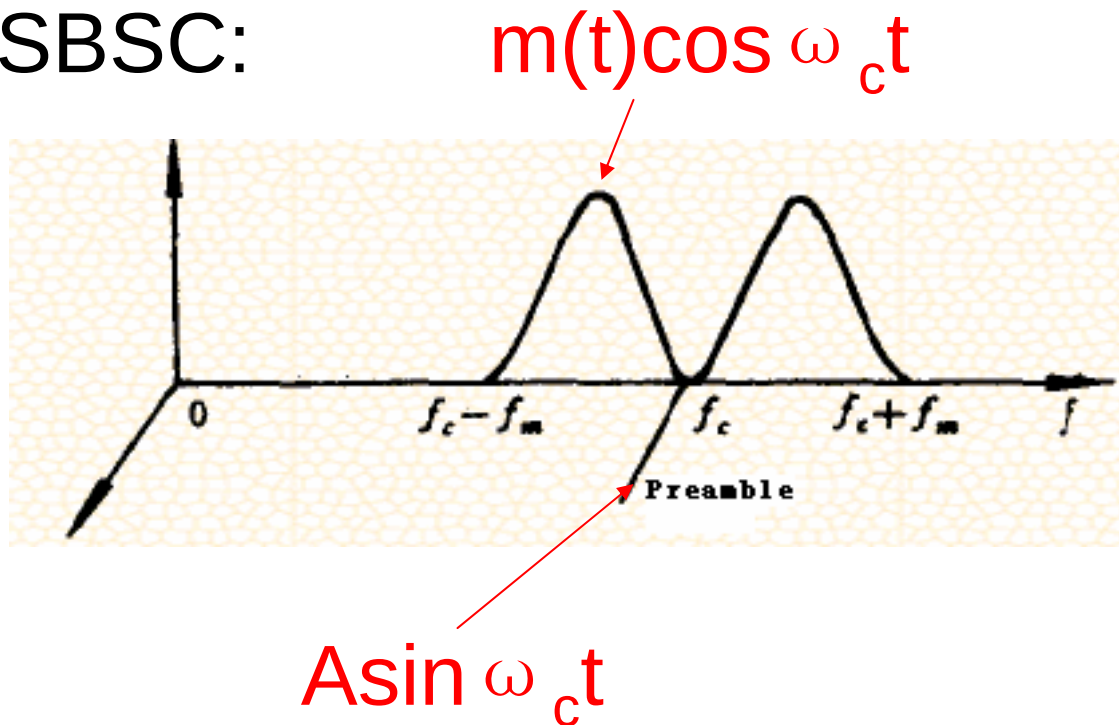
has a **preamble** in a TDM manner

- Nondata-aided synchronization

Carrier Synchronization

- Data-aided synchronization

DSBSC:



Carrier Synchronization (2)

- Nondata-aided synchronization

$$s(t) = m(t)\cos\omega_c t$$

After square-law Device:

$$\begin{aligned} e(t) &= m^2(t)\cos^2\omega_c t \\ &= \frac{m^2(t)}{2} + \frac{1}{2}m^2(t)\cos 2\omega_c t \end{aligned}$$

Using PLL



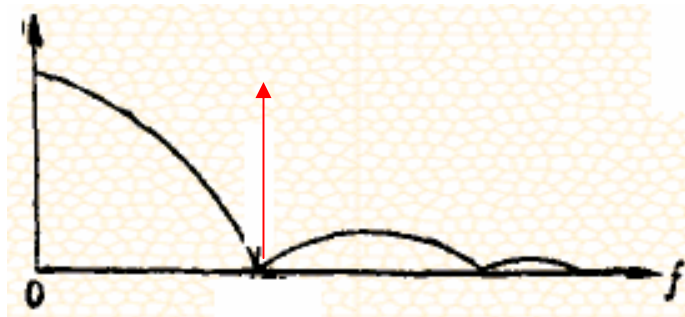
Symbol Synchronization

- Data-aided synchronization

Refer to NRZ

has no discrete frequency

add a preamble f_s

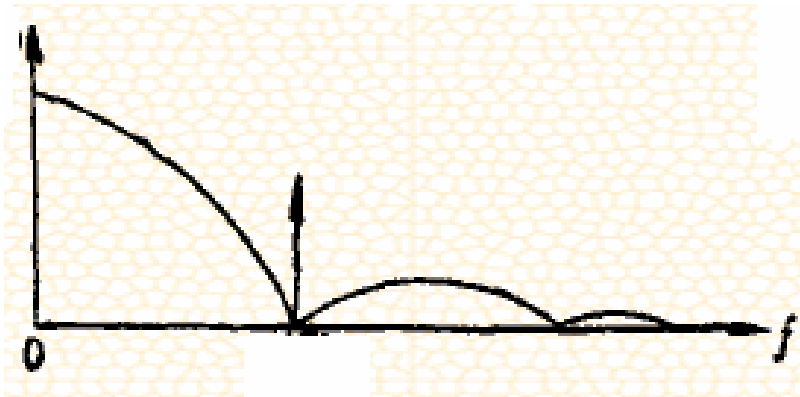


Symbol Synchronization

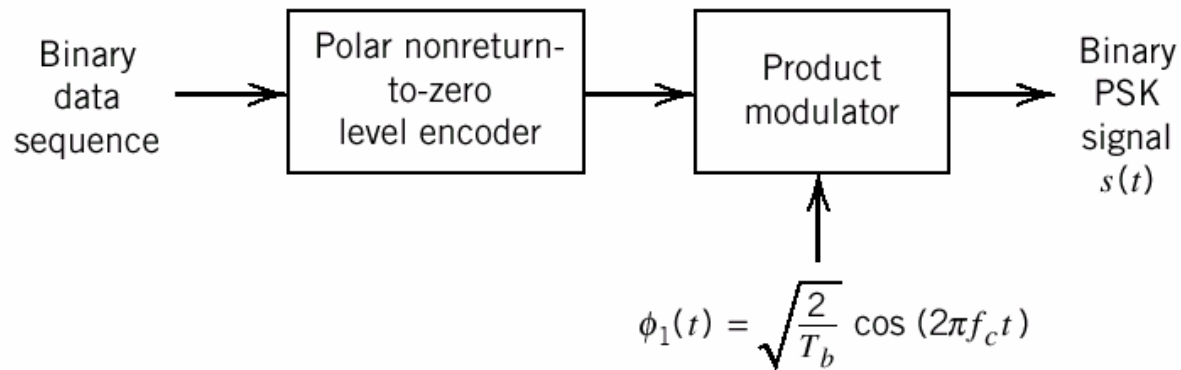
- **Nondata-aided synchronization**

Refer to **RZ**

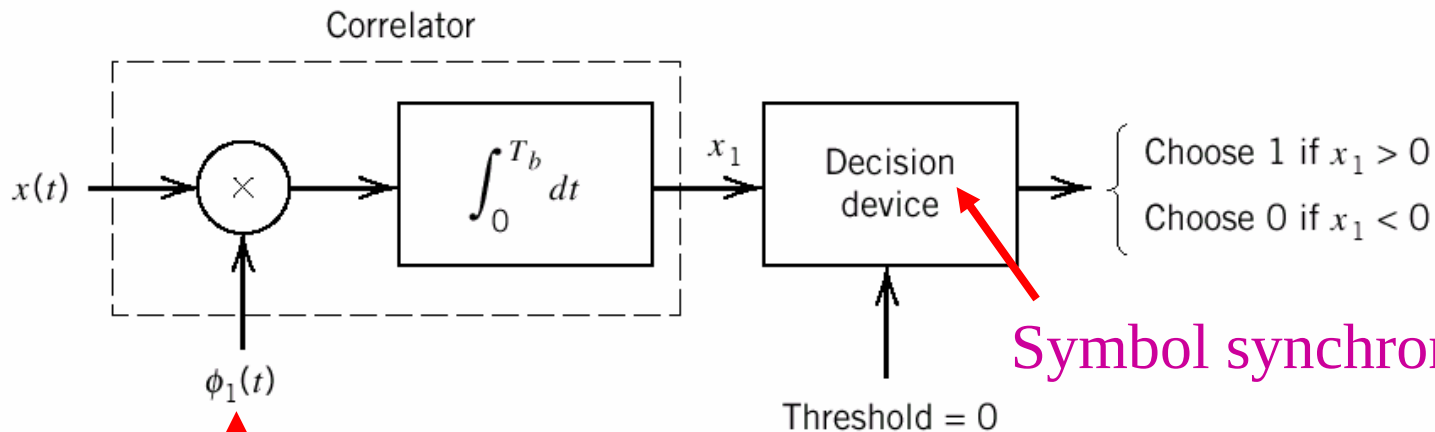
has discrete frequency $\rightarrow$
synchronization component



Summary



(a)



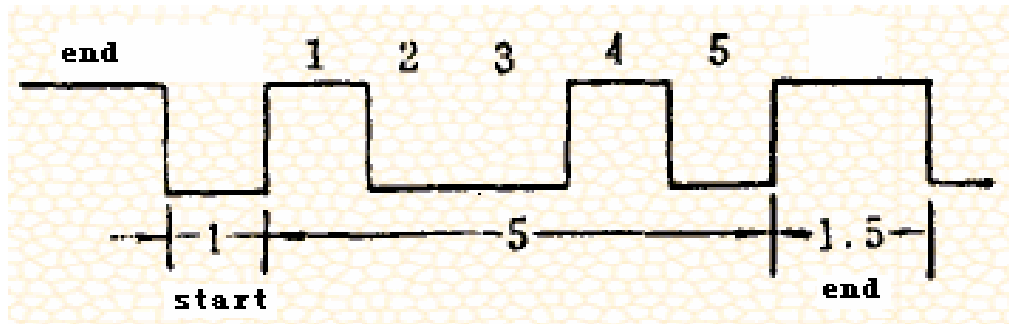
Symbol synchronization

Carrier synchronization

(b)

Frame Synchronization

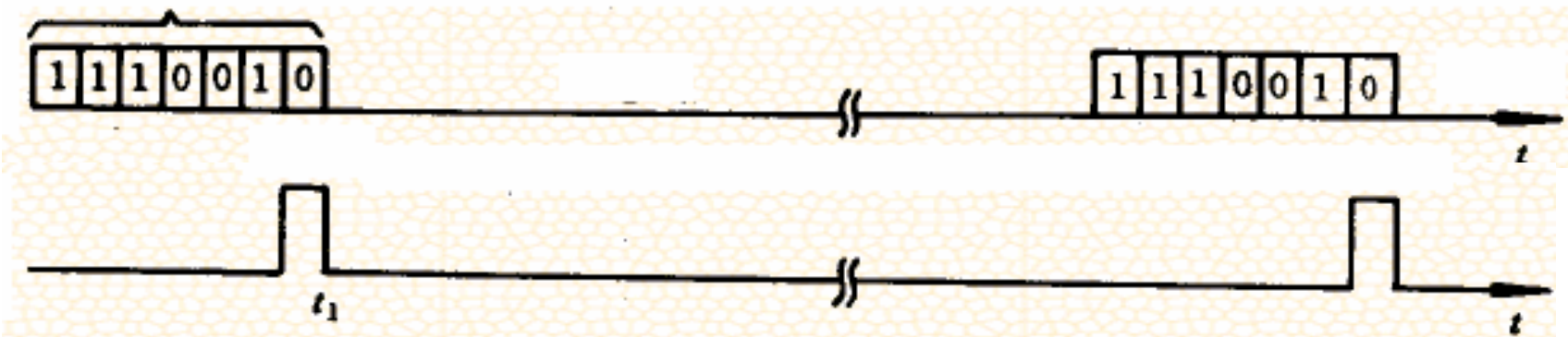
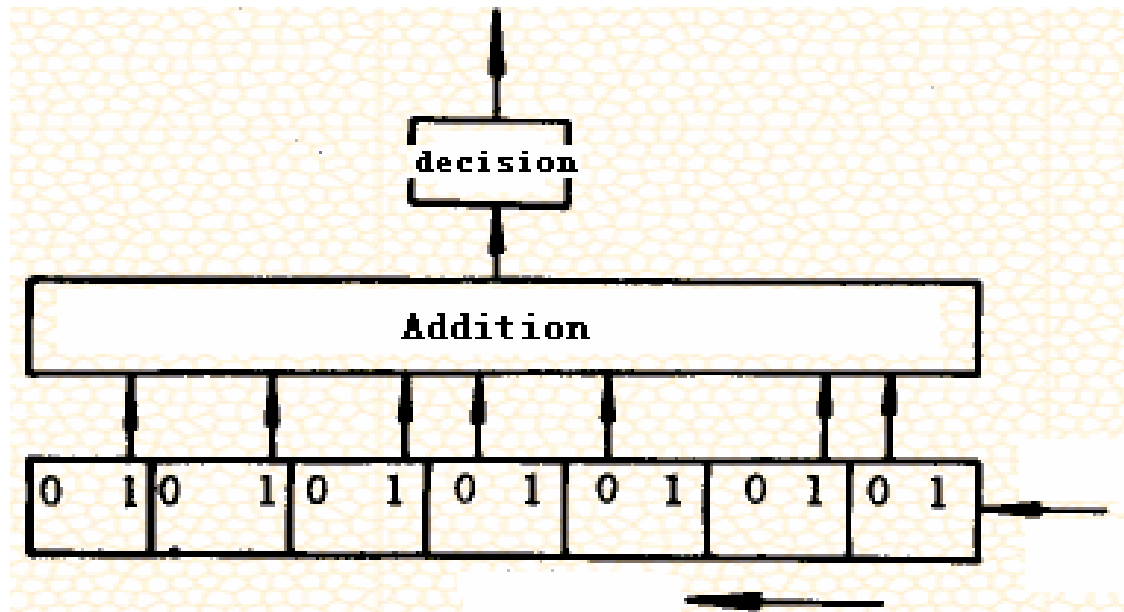
- **Symbol synchronization: word**
- **Frame Synchronization: sentence**
- **Method 1:**



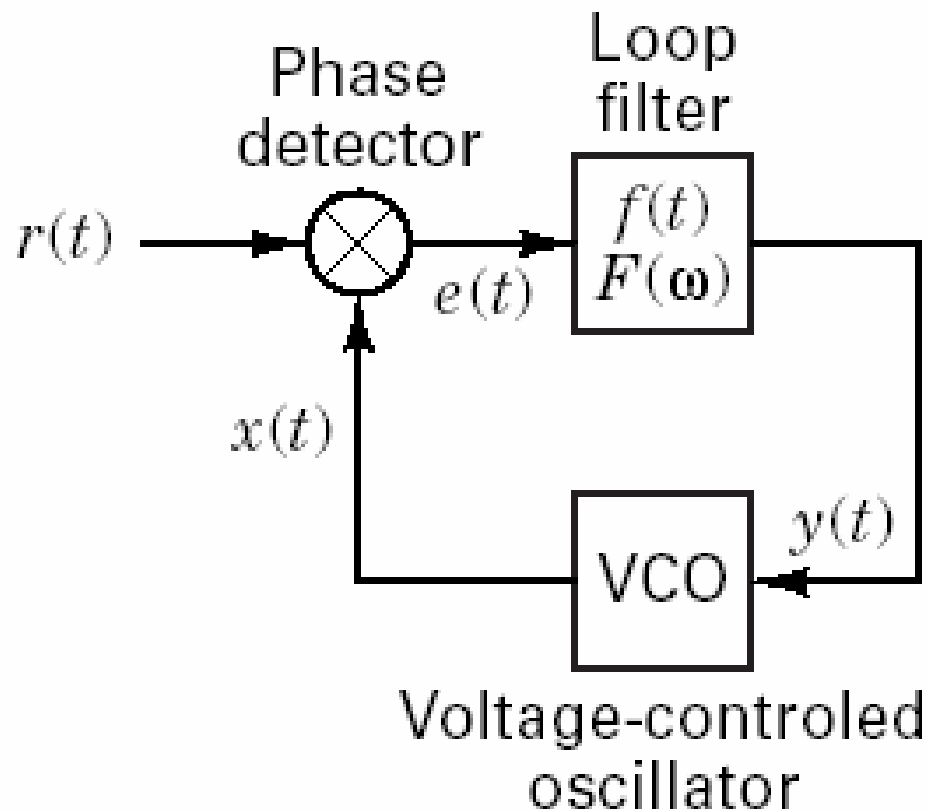
Frame Synchronization (2)

- Method 2:
Barker code

1110010

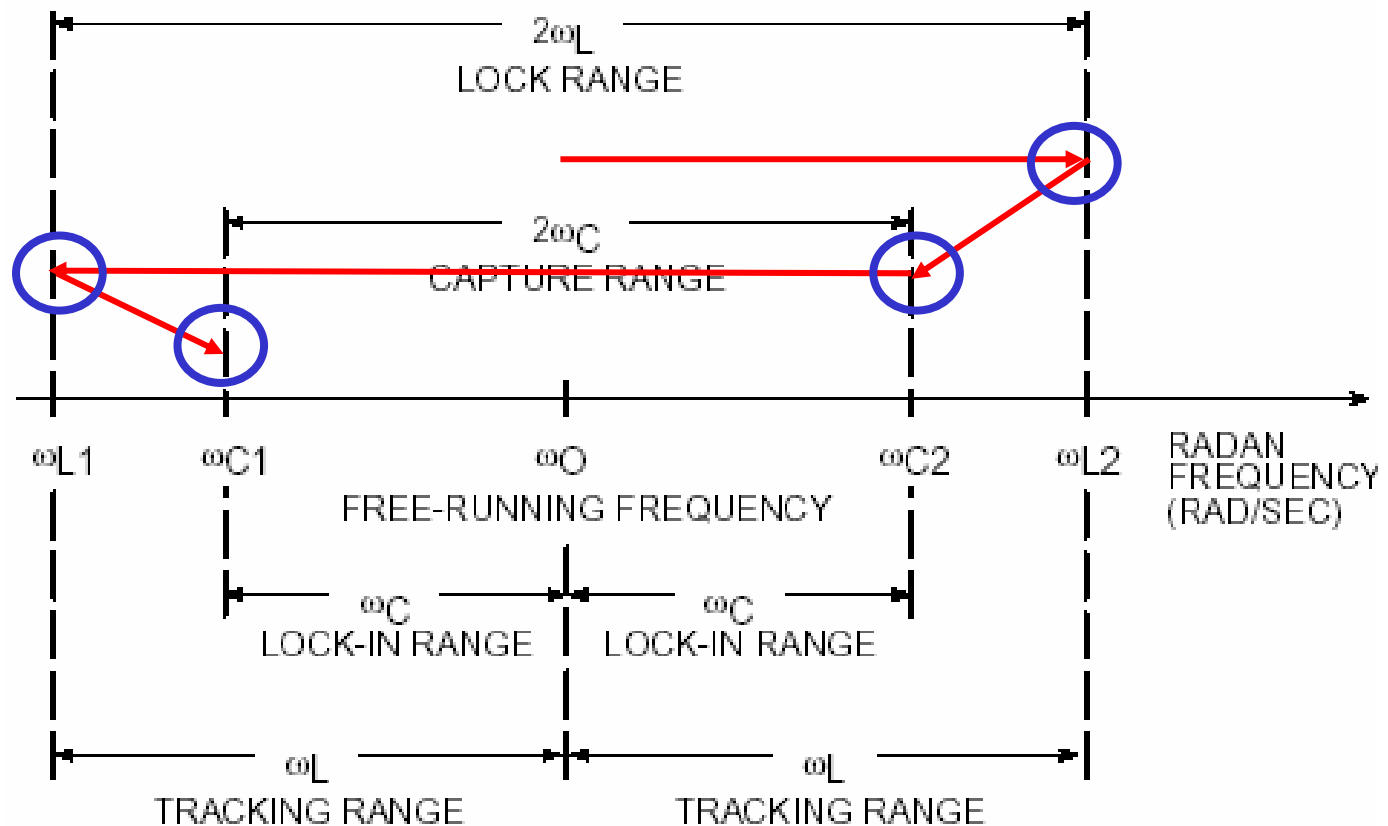


Phase-Locked Loop(PLL)



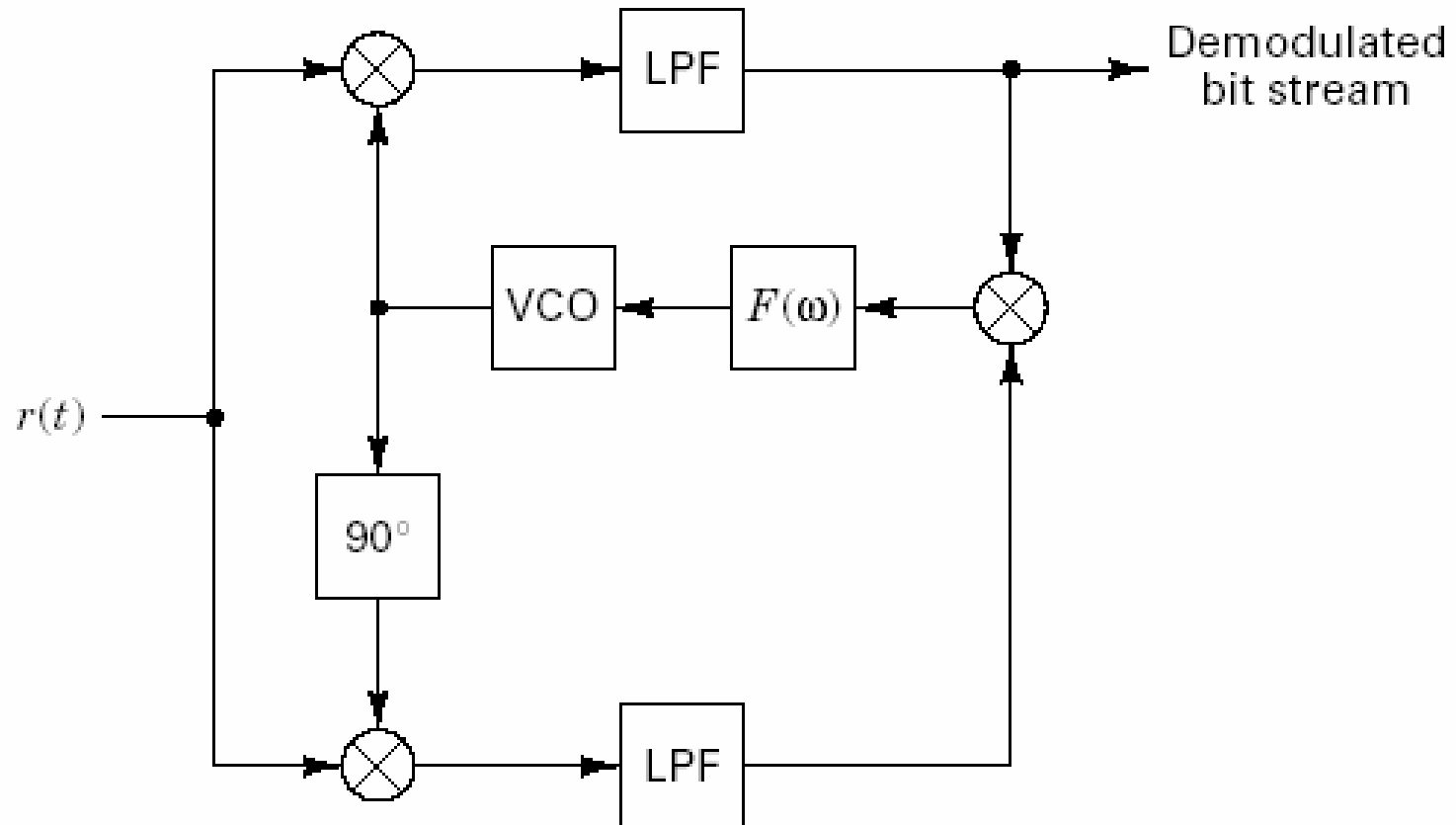
Phase-Locked Loop(PLL)

- Capture Range versus Lock Range



Phase-Locked Loop(e.g.)

- Costas Loop



Q&A
